JCIM JOURNAL OF CHEMICAL INFORMATION AND MODELING

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DO-GMA: An End-to-End Drug−Target Interaction Identification Framework with a Depthwise Overparameterized Convolutional Network and the Gated Multihead Attention Mechanism

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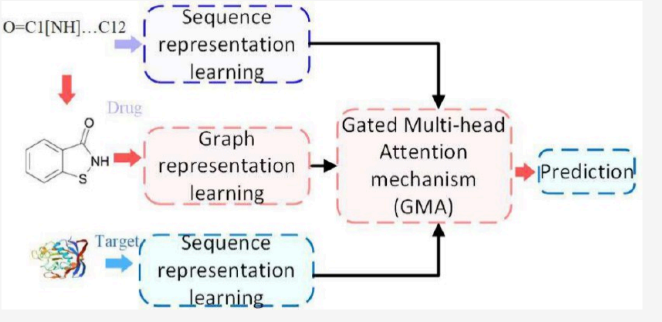
Cite This: J. Chem. Inf. Model. 2025, 65, 1318−1337 Read Online

ABSTRACT: Identification of potential drug−target interactions (DTIs) is a crucial step in drug discovery and repurposing. Although deep learning effectively deciphers DTIs, most deep learning-based methods represent drug features from only a single perspective. Moreover, the fusion method of drug and protein features needs further refinement. To address the above two problems, in this study, we develop a novel end-to-end framework named DO-GMA for potential DTI identification by incorporating Depthwise Overparameterized convolutional neural network and the Gated Multihead Attention mechanism with shared-learned queries and bilinear model concatenation. DO-GMA first designs a depthwise overparameterized convolutional neural network to learn drug representations from their SMILES strings and protein representations from their amino acid sequences. Next, it extracts drug representations from their 2D molecular graphs through a graph convolutional network. Subsequently, it fuses drug and protein features by combining the gated attention mechanism and the multihead attention mechanism with shared-learned queries and bilinear model concatenation. Finally, it takes the fused drug−target features as inputs and builds a multilayer perceptron to classify unlabeled drug−target pairs (DTPs). DO-GMA was benchmarked against six newest DTI prediction methods (CPI-GNN, BACPI, CPGL, DrugBAN, BINDTI, and FOTF-CPI) under four different experimental settings on four DTI data sets (i.e., DrugBank, BioSNAP, C.elegans, and BindingDB). The results show that DO-GMA significantly outperformed the above six methods based on AUC, AUPR, accuracy, F1-score, and MCC. An ablation study, robust statistical analysis, sensitivity analysis of parameters, visualization of the fused features, computational cost analysis, and case analysis further validated the powerful DTI identification performance of DO-GMA. In addition, DO-GMA predicted that two drug−protein pairs (i.e., DB00568 and P06276, and DB09118 and Q9UQD0) could be interacting. DO-GMA is freely available at https://github.com/plhhnu/DO-GMA.



1. INTRODUCTION

Drug−target interaction (DTI) prediction exerts a pivotal role in drug discovery.[1−4] Drugs bind with targets and influence their downstream activity by interactions.[5] Thus, the exploration of potential linkages between medications and targets facilitates better discernment of the pathophysiological characteristics of targeted drugs and therapeutic targets, thereby promoting disease's early prognosis, detection, and therapy.[6]

In vitro experiments are reliable but have a high cost and are time-consuming when screening new DTIs, thereby limiting their large-scale applications. In contrast, in silico methods facilitate narrowing down the screening scope of drug candidates and provide insights into the discovery of new therapeutic clues.[1] Therefore, in silico methods have acquired significant progress in DTI prediction. Traditional computational methods for DTI prediction include molecular docking[7−9] and pharmacophore modeling.[10,11]

Molecular docking evaluates the binding ability between a ligand and a receptor. Hossain et al.[12] analyzed the correlation between physicochemical features of ligands and binding capacity to receptors using AutoDock. Junaidin et al.[8] used DOCK6 and iGemdock to predict compound-drug binding affinity. He et al.[9] revealed therapeutic effects of curcumin on colon cancer by combining the network method and molecular docking. TarFisDock[7] identified possible DTIs through molecular docking.

Received: November 14, 2024
Revised: January 11, 2025
Accepted: January 13, 2025
Published: January 28, 2025

Pharmacophore modeling aims to infer whether a new compound has similar structural features to the observed active compounds.[10] For example, by pharmacophore modeling, Babu et al.[10] inferred potential lead compounds for JAK1. Shareef et al.[13] confirmed that anileridine was one potential BACE-1 inhibitor. Estrada et al.[11] built a 3D structure model to find new inhibitors for human SGLT2.

Although conventional in silico techniques have obtained significant progress, 3D structures of most target proteins are unavailable, thereby limiting their wide applications.[14] In addition, drug-based approaches may result in subpar predictions when a target protein has only a few observed binding molecules.[6]

With the advances of artificial intelligence, machine learning has been broadly applied to bioinformatics including various interaction prediction tasks.[15−17] Since deep learning exhibits a powerful characterization ability in solving complex and high-dimensional data,[14,18] it obtains optimal performance when predicting potential DTIs.[19,20] Deep learning methods for DTI prediction mainly contain three categories: graph-based, transformer-based, and attention-based methods.

Graph-based methods devise graph-based deep learning models to obtain drug representations. For example, to learn drug feature representations, CPI-GNN,[19] GSATDTA,[21] AttenSyn,[22] DRAGNN,[23] and MRLHGNN[24] used a graph neural network (GNN), and CPGL[25] adopted a graph attention network (GAT). IGT[26] and GraphormerDTI[27] employed a graph transformer, and DTIGCCN,[28] GADTI,[29] TGCNDR,[30] and GLCN-DTA[31] applied a graph convolutional network (GCN). However, graph-based methods may perform poorly when connections between DTP nodes are few, especially for isolated nodes.[32] Moreover, graph-based methods require considerable computational resources and are time-consuming because their constructed adjacency matrix is large-scale and may contain millions of nodes.[33] In addition, many graph-based methods usually directly use the similarity matrices of drugs and targets as inputs rather than process their original biological characteristics for interaction prediction.[34]

Transformer-based methods use transformers to encode features of drugs and proteins. MolTrans[35] and Multi-Trans-DTI[36] encoded protein features through the transformer module. FOTF-CPI[37] improved a transformer for predictions. DTITR[38] devised two parallel transformer-encoders to acquire a contextual embedding corresponding to protein sequences and drug SMILES strings. FastDTI[39] explored a RoBERTa architecture-based transformer model to learn drug features. Transformer-based methods severely depend on positional information when encoding.[40] Moreover, they are computationally complex and have high computational requirements.[41]

Attention-based methods learn short- and long-term context correlations between each drug−target pair based on attention mechanisms. For example, to integrate features of drugs and proteins, MRBDTA[42] integrated multihead attention and skip connection. BACPI[43] used the bidirectional attention mechanism. MFR-DTA[44] developed a spatial attention module to acquire the attention-weighted feature map. Perceiver CPI[45] designed a cross-attention model. MCANet[46] used multihead-cross-attention network. CCL-DTI[20] explored the two-sided attention mechanism. DRGBCN[47] adopted a bilinear attention network. MIDTI[48] incorporated a multiview similarity network and the deep interactive attention mechanism. BINDTI[49] combined intention and multihead attention, and Dti-mvsca[50] incorporated the self-attention mechanism and SHADOW graph attention network. Attention-based methods allow one to efficiently incorporate diverse data sources for predicting DTIs more accurately and comprehensively.[51] They still facilitate the reduction of computational burden for large-scale DTI data compared to graph-based and transformer-based methods.[32]

Currently, most DTI prediction methods effectively learn features of drugs through GCN or SMILES sequence-based convolution,[25,37,43,52] but learning drug representations from only a single perspective could be insufficient to comprehensively depict drug features.[51] Moreover, a feature fusion method for drugs and proteins is needed for further improvement.[51] To solve the above problems, here, we consider the advantages of attention-based methods in computational efficiency and data integration and develop DO-GMA, an attention-based end-to-end framework to identify potential DTIs. This study has the following three main contributions:

- An end-to-end framework named DO-GMA is developed to predict potential DTIs by incorporating a Depthwise Overparameterized Convolutional neural network (DO-Conv) and the Gated Multihead Attention mechanism.
- The DO-Conv module is adopted to generate a composite convolution kernel and enhance sequence feature learning of drugs and proteins by incorporating depthwise convolution and conventional convolution.
- A multimodel attention feature fusion module, which incorporates gated attention and multihead attention with shared-learned queries and bilinear model concatenation, is explored to effectively aggregate the features of drugs and targets.

2. MATERIALS AND METHODS

2.1. Data Sets. The DO-GMA performance is evaluated on four publicly available DTI data sets, i.e., DrugBank,[53] BioSNAP,[54] C.elegans,[55] and BindingDB.[56] Their detailed information is listed in Table 1. Particularly, C.elegans[55] includes highly plausible negative and positive DTIs. BindingDB[56] provides extensive binding affinity information and experimentally validated small molecule−protein interactions.

Table 1. Information of Four DTI Data Sets

data set	drug	protein	interaction	positive	negative
DrugBank	6636	4254	34992	17501	17491
BioSNAP	4505	2181	27464	13830	13634
C.elegans	1767	1876	7785	3893	3892
BindingDB	14643	2623	49199	20674	28525

2.2. DO-GMA. As shown in Figure 1, we developed an end-to-end framework called DO-GMA to predict DTI candidates. The framework consists of four main steps. (1) Drug representation learning. Drug features are extracted based on their SMILES strings and 2D molecular graphs by combining DO-Conv and GCN. (2) Protein representation learning. Protein features are learned from their amino acid sequences by DO-Conv. (3) Feature fusion. Drug and protein features are fused by a gated multihead attention (GMA) mechanism. (4) Drug−target pair (DTP) classification. DTPs are classified as interacting or noninteracting by a multilayer perceptron (MLP).

2.2.1. Problem Formulation. For a DTI data set $D = \{(\mathcal{G}_i, \mathcal{P}_i, y_i)\}^m$, $\mathcal{G}_i$ and $\mathcal{P}_i$ denote the SMILES string of the ith drug and amino acid sequence of the ith target protein,

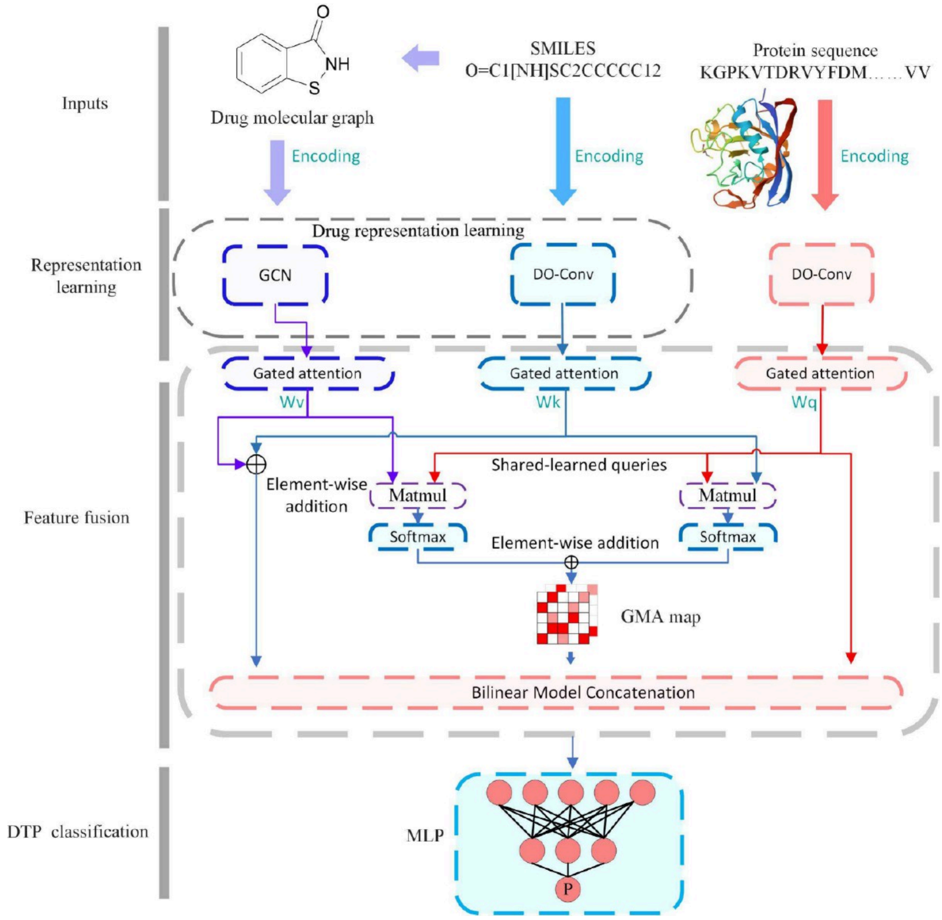


Figure 1. Illustration of the proposed DO-GMA framework for DTI prediction. (1) Drug representation learning. (2) Protein representation learning. (3) Feature fusion. (4) DTP classification.

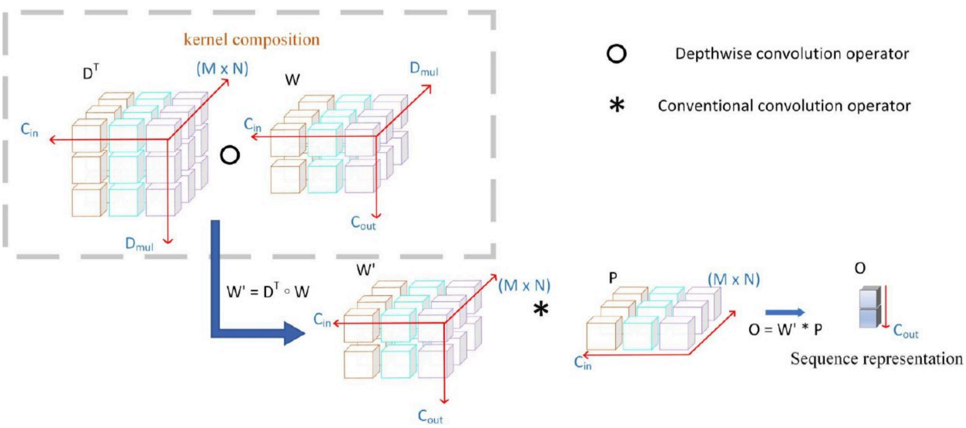


Figure 2. Encoding drug and protein sequence features with DO-Conv.

respectively. y_i denotes the interacting or noninteracting relationship between the ith drug and the ith target protein. Let $\mathbf{feat}_i \in \mathbb{R}^{d_f}$ denote the d_f-dimensional feature vector of the ith DTP. We aim to learn a classifier $F(\mathbf{feat}_i)$ to determine whether the DTP is interacting or noninteracting by eq 1:

$$F(\mathbf{feat}_i) = \begin{cases} 1, & \text{if there is an interaction between } \mathcal{G}_i \text{ and } \mathcal{P}_i \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

The DTP classification model is described by eq 2:

$$F(\mathbf{feat}_i) = \mathrm{MLP}(\mathrm{GMA}(\mathrm{GCN}(\mathcal{G}_i), \text{DO-Conv}(\mathcal{G}_i), \text{DO-Conv}(\mathcal{P}_i))) \tag{2}$$

where MLP(...), GMA(...), GCN($\mathcal{G}_i$), DO-Conv($\mathcal{G}_i$), and DO-Conv($\mathcal{P}_i$) denote the DTP classification results by an MLP, the fused features of drugs and targets obtained by the gated multihead attention mechanism, drug molecular graph representations learned by GCN, drug SMILES string representations extracted by DO-Conv, and protein sequence representations obtained by DO-Conv, respectively.

https://doi.org/10.1021/acs.jcim.4c02088
J. Chem. Inf. Model. 2025, 65, 1318−1337

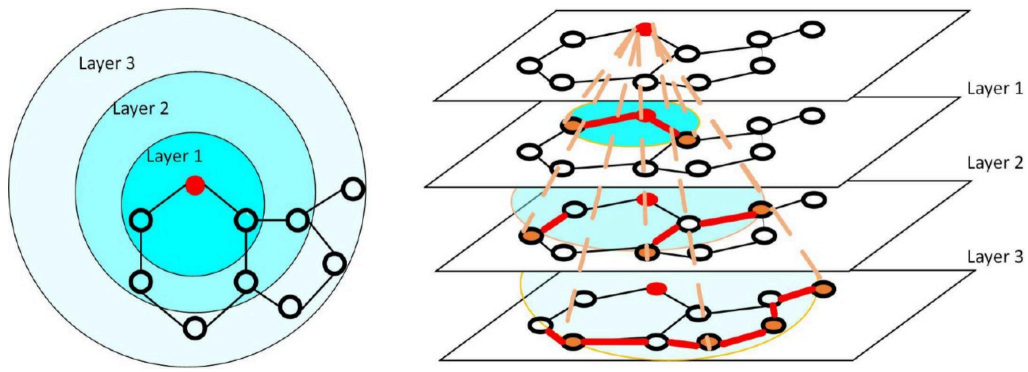


Figure 3. Learning drug molecular graph representation by GCN.

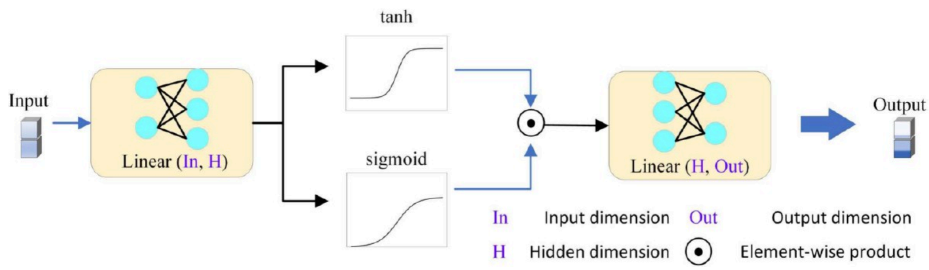


Figure 4. Gated attention mechanism module.

2.2.2. Encoding Drug and Protein Sequence Features with DO-Conv. The SMILES strings provide rich and complex sequence information for drugs.[52] To improve DTP classification performance, as shown in Figure 2, we propose a DO-Conv module to precisely learn drug features from their SMILES strings[57] by incorporating depthwise convolution and traditional convolution.

Specially, each SMILES string is first transformed to a patch $\boldsymbol{P} \in \mathbb{R}^{(M\times N)\times C_{in}}$ with C_{in} channels and two spatial dimensions M and N, where M and N are computed according to the size of the same convolutional kernel. Next, the DO-Conv module takes the obtained patches as inputs and outputs a C_{out}-dimensional (i.e., C_{out} channels) vector $\boldsymbol{O} \in \mathbb{R}^{C_{out}}$ to represent each drug by incorporating the depthwise convolution module with the trainable kernel $\boldsymbol{D} \in \mathbb{R}^{(M\times N)\times D_{mul}\times C_{in}}$ (which contains D_{mul} depth multipliers) and the conventional convolution module with the trainable kernel $\boldsymbol{W} \in \mathbb{R}^{C_{out}\times D_{mul}\times C_{in}}$.

Specifically, the depthwise convolution module first adopts a depthwise convolution operator ∘ to generate a composite kernel $\boldsymbol{W}' \in \mathbb{R}^{C_{out}\times(M\times N)\times C_{in}}$ based on two trainable kernels $\boldsymbol{W}$ and $\boldsymbol{D}$ by eq 3:

$$\boldsymbol{W}'_{c_{out}jc_{in}} = \boldsymbol{D}^{T}_{\alpha jc_{in}} \circ \boldsymbol{W}_{c_{out}\alpha c_{in}} \tag{3}$$

where $\boldsymbol{D}^T \in \mathbb{R}^{D_{mul}\times(M\times N)\times C_{in}}$ denotes the transpose of $\boldsymbol{D}$, and c_{out}, α, j, and c_{in} donate different channels.

Next, the conventional convolution module uses a convolution operator * to obtain drug SMILES representations $\boldsymbol{O}$ based on $\boldsymbol{W}'$ and $\boldsymbol{P}$ by eq 4:

$$\boldsymbol{v}_{ds} = \boldsymbol{O}_{c_{out}} = \boldsymbol{W}'_{c_{out}\gamma} * \boldsymbol{P}_{\gamma} \tag{4}$$

In summary, the DO-Conv module generates a composite convolution kernel by integrating depthwise convolution and conventional convolution. Consequently, it increases the learnable parameters during training and further promotes the model generalization ability. Moreover, operators in the two convolutions can be independently computed, thereby enhancing the model computational efficiency and interpretability.

Similarly, for a protein, we learn its representation $\boldsymbol{v}_p$ based on its amino acid sequence through the DO-Conv module.

2.2.3. Encoding Drug Molecular Graph Features with GCN. Drug molecular graphs are chemical structures containing entities (i.e., atoms) and relationships (i.e., bonds) between atoms. These molecular graphs provide abundant semantic information and complex spatial structures. GCN demonstrates powerful feature learning capability from unstructured data, especially graphs, by introducing a convolution operation. Here, to make atoms aggregate information from neighbor nodes, as shown in Figure 3, we adopt the Python package DGL-LifeSci[58] to convert drug SMILES strings to a two-dimensional molecular graph based on their chemical properties and then update atom features through three graph convolutional layers. Consequently, drug molecular graph features with C_{out} embedding size are encoded through GCN by eq 5:

$$\boldsymbol{H}^{(l+1)} = \sigma(\boldsymbol{D}^{-1/2}\boldsymbol{A}\boldsymbol{D}^{-1/2}\boldsymbol{H}^{(l)}\boldsymbol{Weight}^{(l)}) \tag{5}$$

where $\boldsymbol{H}^{(l)} \in \mathbb{R}^{N_d\times C_{out}}$, $\boldsymbol{A} \in \mathbb{R}^{N_d\times N_d}$, $\boldsymbol{D} \in \mathbb{R}^{N_d\times N_d}$, $\boldsymbol{Weight}^{(l)} \in \mathbb{R}^{C_{out}\times C_{out}}$, and σ denote the drug molecular graph feature matrix at the lth layer, weighted adjacency matrix, the diagonal matrix of $\boldsymbol{A}$, weight matrix at the lth layer, and the ReLU activation function, respectively. N_d is a maximum number of atoms and is used to convert each drug into a tensor with the same dimension. Since there are different atom numbers for different drugs, we pad the virtual nodes with zero for the drugs with less atoms. Finally, the drug molecular graph is depicted as $\boldsymbol{v}_{dg} = \boldsymbol{H}^{(3)} \in \mathbb{R}^{N_d\times C_{out}}$ when the layer number of GCN is set to 3.

2.2.4. Fusing Drug and Target Features with GMA. The multihead attention mechanism is usually introduced to capture the correlations between different feature spaces, obtain attention weight mapping, and complete feature interactions.

https://doi.org/10.1021/acs.jcim.4c02088
J. Chem. Inf. Model. 2025, 65, 1318−1337

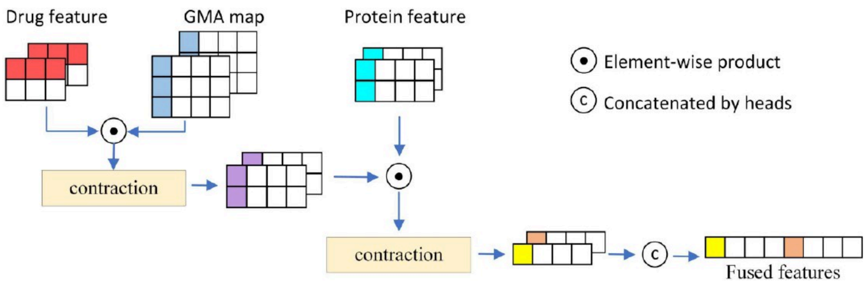


Figure 5. Bilinear model concatenation module.

The traditional methods improve the performance of the multihead attention mechanism by adding channels or attention heads, thereby resulting in a larger memory consumption and computational burden. During DTP representation learning, to reduce computational resource requirements while promoting feature fusion of drug and proteins, we present a novel gated multihead attention (GMA) mechanism with shared-learned queries and bilinear model concatenation based on the obtained drug SMILES representations, drug molecular graph representations, and protein sequence representations.

The GMA mechanism contains three modules: (1) the feature enhancement module based on the gated attention mechanism, (2) the multihead attention module with shared-learned queries for computing attention scores between drugs and targets, and (3) the bilinear model concatenation module, which takes the enhanced representations and the multihead attention scores as inputs and produces the fused DTP features.

The gating mechanism can potentially alleviate the troublesome linear nature in data through a learnable nonlinearity mechanism.[59] As shown in Figure 4, the gated attention mechanism[60] with d_h hidden dimensions is used as the feature enhancement module $s(\cdot)$ to enhance features of drugs and proteins by eq 6:

$$\begin{cases} s(\boldsymbol{H}_g) = (\tanh(\boldsymbol{H}_g\boldsymbol{V}) \odot \mathrm{Sigmoid}(\boldsymbol{H}_g\boldsymbol{U}))\boldsymbol{w}_a \\ \mathbf{M}_{\mathbf{ds}} = s(\boldsymbol{v}_{ds}) \\ \mathbf{M}_{\mathbf{dg}} = s(\boldsymbol{v}_{dg}) \\ \mathbf{M}_{\mathbf{p}} = s(\boldsymbol{v}_{p}) \end{cases} \tag{6}$$

where $\boldsymbol{w}_a \in \mathbb{R}^{d_h \times d_f}$, $\boldsymbol{V} \in \mathbb{R}^{C_{out} \times d_h}$, and $\boldsymbol{U} \in \mathbb{R}^{C_{out} \times d_h}$ are learnable matrices, $\boldsymbol{H}_g$ is the instances to be enhanced, and $\boldsymbol{v}_{ds}$, $\boldsymbol{v}_{dg}$, and $\boldsymbol{v}_p$ are drug sequence representations, drug molecular graph representations, and protein sequence representations, respectively. Sigmoid$(\cdot)$ and tanh$(\cdot)$ denote the sigmoid and tanh functions, respectively. $\odot$ is the element-wise product.

Furthermore, the shared-learned queries across windows effectively leverage the weight-sharing principle and the agent attention and fully utilize redundancy between attention weights.[61,62] Thus, it acquires better model expressiveness and less computational complexity, expedites the generation process, and tremendously promotes generation quality. Here, we build two multihead attention score modules with shared-learned queries ($\mathbf{M}_\mathbf{p}\mathbf{W}_q^T$) to improve the layer expressiveness and compute attention weights $\boldsymbol{att\_map}_{dg}$ and $\boldsymbol{att\_map}_{ds}$ by eq 7:

$$\begin{cases} \boldsymbol{att\_map}_{ds} = \mathrm{Softmax}\left(\dfrac{(\mathbf{M}_\mathbf{p}\boldsymbol{W}_q{}^T)(\mathbf{M}_{\mathbf{ds}}\boldsymbol{W}_k{}^T)^T}{\sqrt{d_k}}\right) \\ \boldsymbol{att\_map}_{dg} = \mathrm{Softmax}\left(\dfrac{(\mathbf{M}_\mathbf{p}\boldsymbol{W}_q{}^T)(\mathbf{M}_{\mathbf{dg}}\boldsymbol{W}_v{}^T)^T}{\sqrt{d_k}}\right) \end{cases} \tag{7}$$

where $\boldsymbol{att\_map}_{dg}$ and $\boldsymbol{att\_map}_{ds}$ have the same dimension d_k. $\boldsymbol{W}_q \in \mathbb{R}^{d_f \times d_f}$, $\boldsymbol{W}_k \in \mathbb{R}^{d_f \times d_f}$, and $\boldsymbol{W}_v \in \mathbb{R}^{d_f \times d_f}$ are new learnable parameters. $\boldsymbol{W}_q^T$, $\boldsymbol{W}_k^T$, and $\boldsymbol{W}_v^T$ denote the transpose of $\boldsymbol{W}_q$, $\boldsymbol{W}_k$, and $\boldsymbol{W}_v$, respectively.

Consequently, we use the element-wise addition operation to obtain the entire drug features $\mathbf{v\_d}$ and protein features $\mathbf{v\_p}$ by eq 8:

$$\begin{cases} \mathbf{v\_d} = \mathbf{M}_{\mathbf{ds}}\mathbf{W}_k{}^T + \mathbf{M}_{\mathbf{dg}}\mathbf{W}_v{}^T \\ \mathbf{v\_p} = \mathbf{M}_\mathbf{p}\mathbf{W}_q{}^T \end{cases} \tag{8}$$

where $\mathbf{v\_d} \in \mathbb{R}^{S \times K}$ and $\mathbf{v\_p} \in \mathbb{R}^{Q \times K}$ with S and Q representing different number of vertices and K representing the dimension in embedding space. $\boldsymbol{W}_q$, $\boldsymbol{W}_k$, and $\boldsymbol{W}_v$ are shared with the previous interactive map layer, thus decreasing the number of parameters and alleviating overfitting.

Next, we use the element-wise addition operation to fuse the attention weights with shared-learned queries to a composite attention weight $\boldsymbol{att\_map}$ for each attention head by eq 9:

$$\boldsymbol{att\_map} = \boldsymbol{att\_map}_{ds} + \boldsymbol{att\_map}_{dg} \tag{9}$$

The bilinear model investigates every pair of features and utilizes a quadratic expansion of linear transformation.[63] Thus, it can aggregate multiple features, decipher associations between different features, and alleviate the curse of dimensionality. The bilinear model maps two different types of vector spaces $\mathbf{x}$ and $\mathbf{y}$ to a new observation space $\boldsymbol{z}$ based on a weight matrix $\boldsymbol{A}$ by eq 10:

$$\boldsymbol{z} = \mathbf{x}^T\boldsymbol{A}\mathbf{y} \tag{10}$$

Here, inspired by the bilinear pooling technique,[52] we devise a novel module named bilinear model concatenation to fuse drug features $\mathbf{v\_d} \in \mathbb{R}^{S \times K}$ and protein features $\mathbf{v\_p} \in \mathbb{R}^{Q \times K}$. As shown in Figure 5, the bilinear model concatenation module combines attention weights $\boldsymbol{att\_map} \in \mathbb{R}^{Q \times S}$ obtained from the multihead attention mechanism and drug representations and protein representations, making the model effectively distinguish feature importance and avoid overfitting. For each attention head, to begin with, we perform element-wise product and summation on $\mathbf{v\_d}$ and $\boldsymbol{att\_map}$ to obtain a tensor along the dimension S by eq 11:

https://doi.org/10.1021/acs.jcim.4c02088
J. Chem. Inf. Model. 2025, 65, 1318−1337

Table 2. Performance Comparison under E_1 to E_4 on DrugBank (Running 10 Random Experiments)

method	AUC	AUPR	accuracy	F1-score	MCC
			Setting E_1		
CPI-GNN	0.6846 ± 0.0052	0.7283 ± 0.0050	0.6439 ± 0.0079	0.6214 ± 0.0073	0.2908 ± 0.0172
BACPI	0.8576 ± 0.0066	0.8672 ± 0.0073	0.7773 ± 0.0081	0.7768 ± 0.0098	0.5554 ± 0.0165
CPGL	0.8531 ± 0.0050	0.8578 ± 0.0064	0.7732 ± 0.0067	0.7776 ± 0.0069	0.5470 ± 0.0130
DrugBAN	0.8783 ± 0.0048	0.8777 ± 0.0087	0.8091 ± 0.0065	0.8110 ± 0.0056	0.6184 ± 0.0128
BINDTI	0.8801 ± 0.0052	0.8790 ± 0.0072	0.8106 ± 0.0062	0.8117 ± 0.0046	0.6215 ± 0.0123
FOTF-CPI	0.8751 ± 0.0047	0.8792 ± 0.0051	0.7990 ± 0.0064	0.8024 ± 0.0061	0.5983 ± 0.0125
DO-GMA	**0.9095 ± 0.0036**	**0.9143 ± 0.0041**	**0.8312 ± 0.0052**	**0.8345 ± 0.0051**	**0.6629 ± 0.0103**
			Setting E_2		
CPI-GNN	0.5380 ± 0.0098	0.5410 ± 0.0135	0.5231 ± 0.0096	0.4460 ± 0.0275	0.0502 ± 0.0187
BACPI	0.7353 ± 0.0086	0.7503 ± 0.0067	0.6639 ± 0.0127	0.6069 ± 0.0444	0.3457 ± 0.0139
CPGL	0.7436 ± 0.0100	0.7496 ± 0.0106	0.6727 ± 0.0103	0.6283 ± 0.0331	0.3581 ± 0.0149
DrugBAN	0.7566 ± 0.0095	0.7738 ± 0.0103	0.6714 ± 0.0182	0.7063 ± 0.0073	0.3530 ± 0.0313
BINDTI	0.7590 ± 0.0068	0.7655 ± 0.0094	0.6704 ± 0.0139	0.7120 ± 0.0056	0.3557 ± 0.0224
FOTF-CPI	0.7499 ± 0.0080	0.7666 ± 0.0089	0.6775 ± 0.0131	0.6127 ± 0.0424	0.3802 ± 0.0142
DO-GMA	**0.7967 ± 0.0062**	**0.8098 ± 0.0066**	**0.6996 ± 0.0124**	**0.7339 ± 0.0062**	**0.4137 ± 0.0202**
			Setting E_3		
CPI-GNN	0.5827 ± 0.0117	0.6188 ± 0.0125	0.5671 ± 0.0075	0.4698 ± 0.0240	0.1444 ± 0.0186
BACPI	0.5988 ± 0.0112	0.6304 ± 0.0116	0.5792 ± 0.0070	0.4435 ± 0.0202	0.1816 ± 0.0176
CPGL	0.6036 ± 0.0109	0.6155 ± 0.0106	0.5683 ± 0.0071	0.4374 ± 0.0219	0.1539 ± 0.0168
DrugBAN	0.5769 ± 0.0076	0.6023 ± 0.0112	0.5014 ± 0.0036	0.6671 ± 0.0005	0.0251 ± 0.0204
BINDTI	0.6014 ± 0.0208	0.6197 ± 0.0216	0.5116 ± 0.0202	0.6685 ± 0.0022	0.0638 ± 0.0430
FOTF-CPI	0.6059 ± 0.0103	0.6428 ± 0.0101	0.5789 ± 0.0064	0.4173 ± 0.0343	0.1899 ± 0.0176
DO-GMA	**0.6955 ± 0.0118**	**0.6945 ± 0.0160**	**0.6008 ± 0.0230**	**0.6901 ± 0.0063**	**0.2475 ± 0.0326**
			Setting E_4		
CPI-GNN	0.5251 ± 0.0127	0.5237 ± 0.0081	0.5145 ± 0.0097	0.3945 ± 0.0406	0.0346 ± 0.0168
BACPI	0.5334 ± 0.0116	0.5343 ± 0.0111	0.5069 ± 0.0113	0.1391 ± 0.1036	0.0283 ± 0.0317
CPGL	0.5451 ± 0.0085	0.5388 ± 0.0085	0.5052 ± 0.0116	0.1118 ± 0.1227	0.0242 ± 0.0272
DrugBAN	0.5466 ± 0.0171	0.5417 ± 0.0159	0.5038 ± 0.0046	0.6669 ± 0.0002	0.0212 ± 0.0108
BINDTI	0.5409 ± 0.0154	0.5394 ± 0.0109	0.5065 ± 0.0084	0.6674 ± 0.0013	0.0287 ± 0.0312
FOTF-CPI	0.5580 ± 0.0247	0.5482 ± 0.0201	0.5086 ± 0.0075	0.1675 ± 0.1118	0.0398 ± 0.0086
DO-GMA	**0.6003 ± 0.0075**	**0.5751 ± 0.0137**	**0.5389 ± 0.0220**	**0.6737 ± 0.0038**	**0.1264 ± 0.0547**

$$\boldsymbol{temp}_{qk} = \sum_{s}^{S} \mathbf{v\_d}_{sk} \odot \boldsymbol{att\_map}_{qs} \tag{11}$$

where q, k, and s donate different channels.

Subsequently, $\boldsymbol{temp}$ and $\mathbf{v\_p}$ undergo element-wise product and summation along the dimension Q to obtain the fused drug and protein features f' by eq 12:

$$f'_k = \sum_{q}^{Q} \boldsymbol{temp}_{qk} \odot \mathbf{v\_p}_{qk} \tag{12}$$

Finally, the fused drug and protein features are concatenated to obtain the fused DT feature f by eq 13:

$$\boldsymbol{f} = \mathrm{Concat}(\boldsymbol{f}'_1, ..., \boldsymbol{f}'_n) \tag{13}$$

where n denotes the number of attention heads.

As a result, there are no new learnable parameters in the bilinear model concatenation module.

2.2.5. DTP Classification. To classify unknown DTPs, we took the fused DT feature f as input and fed it into an MLP with three fully connected layers. Finally, the interaction probability p for each DTP is computed by eq 14:

$$p = \mathrm{Sigmoid}(\boldsymbol{W}\boldsymbol{f} + b) \tag{14}$$

For a given interaction probability threshold δ, the DTP is considered to be interacting when $p > \delta$, otherwise it is noninteracting. During training, we used cross entropy loss to optimize the model.

3. RESULTS

3.1. Experimental Setups. Evaluation metrics: To assess the DO-GMA performance for DTI prediction, we used accuracy, F1-score, MCC (Matthews correlation coefficient), AUC (area under the receiver operating characteristic (ROC) curve), and AUPR (area under the precision-recall (PR) curve) as metrics, where AUC and AUPR were taken as the major evaluation metrics.

Evaluation strategies: In each experiment, the data set was divided into three parts: training, validation, and test sets. Suppose that D_{train} and P_{train} denote the sets of drugs and proteins in the training set. We employed four different experimental strategies to conduct comprehensive comparisons for inferring the linkage between a drug d and a protein p in the test set:

- E_1: Both drug d and protein p were included in the training set, that is, $d \in D_{\mathrm{train}}$ and $p \in P_{\mathrm{train}}$.
- E_2: Drug d was not included in the training set, $d \notin D_{\mathrm{train}}$, that is, there were no interactions linking to d in the training set. However, protein p was included in the training set, $p \in P_{\mathrm{train}}$, that is, there were interactions linking to p in the training set.
- E_3: Protein p was not included in the training set, $p \notin P_{\mathrm{train}}$, that is, there were no interactions linking to p in the

Table 3. Performance Comparison under E_1 to E_4 on BioSNAP (Running 10 Random Experiments)

method	AUC	AUPR	accuracy	F1-score	MCC
			Setting E_1		
CPI-GNN	0.6968 ± 0.0067	0.7193 ± 0.0067	0.6544 ± 0.0064	0.6519 ± 0.0049	0.3096 ± 0.0134
BACPI	0.8876 ± 0.0036	0.8951 ± 0.0046	0.8079 ± 0.0068	0.8097 ± 0.0061	0.6162 ± 0.0132
CPGL	0.8859 ± 0.0072	0.8927 ± 0.0083	0.8103 ± 0.0079	0.8128 ± 0.0089	0.6208 ± 0.0158
DrugBAN	<u>0.9040 ± 0.0042</u>	<u>0.9074 ± 0.0060</u>	<u>0.8373 ± 0.0068</u>	<u>0.8377 ± 0.0050</u>	<u>0.6750 ± 0.0135</u>
BINDTI	0.9025 ± 0.0022	0.9034 ± 0.0033	0.8317 ± 0.0052	0.8350 ± 0.0031	0.6641 ± 0.0100
FOTF-CPI	0.8990 ± 0.0026	0.9025 ± 0.0039	0.8271 ± 0.0040	0.8290 ± 0.0049	0.6544 ± 0.0078
DO-GMA	**0.9239 ± 0.0026**	**0.9261 ± 0.0050**	**0.8515 ± 0.0027**	**0.8541 ± 0.0022**	**0.7040 ± 0.0047**
			Setting E_2		
CPI-GNN	0.5620 ± 0.0118	0.5633 ± 0.0168	0.5400 ± 0.0108	0.4470 ± 0.0534	0.0886 ± 0.0215
BACPI	0.7805 ± 0.0089	0.7971 ± 0.0084	0.7051 ± 0.0118	0.6630 ± 0.0284	0.4286 ± 0.0148
CPGL	0.7803 ± 0.0116	0.7904 ± 0.0221	0.7129 ± 0.0118	0.6889 ± 0.0173	0.4343 ± 0.0257
DrugBAN	<u>0.7971 ± 0.0076</u>	<u>0.8119 ± 0.0102</u>	0.7167 ± 0.0108	<u>0.7337 ± 0.0079</u>	0.4373 ± 0.0198
BINDTI	0.7913 ± 0.0058	0.8006 ± 0.0130	0.7075 ± 0.0140	0.7312 ± 0.0054	0.4213 ± 0.0239
FOTF-CPI	0.7890 ± 0.0071	0.8042 ± 0.0152	<u>0.7245 ± 0.0098</u>	0.6888 ± 0.0254	<u>0.4671 ± 0.0128</u>
DO-GMA	**0.8280 ± 0.0105**	**0.8457 ± 0.0088**	**0.7459 ± 0.0107**	**0.7550 ± 0.0099**	**0.4936 ± 0.0201**
			Setting E_3		
CPI-GNN	0.5472 ± 0.0143	0.5631 ± 0.0205	0.5405 ± 0.0111	0.4258 ± 0.0267	0.0915 ± 0.0282
BACPI	0.5629 ± 0.0147	0.5811 ± 0.0192	**0.5443 ± 0.0108**	0.3152 ± 0.0795	<u>0.1252 ± 0.0208</u>
CPGL	0.5386 ± 0.0191	0.5450 ± 0.0231	0.5213 ± 0.0100	0.2507 ± 0.0740	0.0711 ± 0.0221
DrugBAN	0.5500 ± 0.0107	0.5494 ± 0.0188	0.5049 ± 0.0074	0.6670 ± 0.0005	0.0209 ± 0.0193
BINDTI	0.5680 ± 0.0162	0.5737 ± 0.0158	0.5113 ± 0.0104	<u>0.6674 ± 0.0005</u>	0.0500 ± 0.0291
FOTF-CPI	<u>0.5932 ± 0.0160</u>	<u>0.6035 ± 0.0133</u>	<u>0.5430 ± 0.0089</u>	0.3111 ± 0.0699	**0.1253 ± 0.0183**
DO-GMA	**0.6258 ± 0.0232**	**0.6210 ± 0.0203**	0.5412 ± 0.0247	**0.6731 ± 0.0057**	0.1244 ± 0.0606
			Setting E_4		
CPI-GNN	0.5050 ± 0.0105	0.5106 ± 0.0125	0.5016 ± 0.0073	0.3513 ± 0.0581	0.0092 ± 0.0133
BACPI	0.5226 ± 0.0152	0.5260 ± 0.0167	0.5005 ± 0.0104	0.0882 ± 0.0672	0.0291 ± 0.0331
CPGL	0.5135 ± 0.0211	0.5131 ± 0.0187	0.4955 ± 0.0109	0.0960 ± 0.1472	0.0047 ± 0.0263
DrugBAN	0.5377 ± 0.0190	0.5278 ± 0.0180	<u>0.5079 ± 0.0074</u>	<u>0.6671 ± 0.0004</u>	<u>0.0294 ± 0.0174</u>
BINDTI	0.5279 ± 0.0153	0.5251 ± 0.0168	0.5067 ± 0.0082	0.6669 ± 0.0003	0.0206 ± 0.0141
FOTF-CPI	<u>0.5686 ± 0.0466</u>	**0.5582 ± 0.0414**	0.5032 ± 0.0134	0.1379 ± 0.1190	0.0275 ± 0.0288
DO-GMA	**0.5792 ± 0.0262**	<u>0.5513 ± 0.0303</u>	**0.5402 ± 0.0153**	**0.6726 ± 0.0030**	**0.1262 ± 0.0369**

training set. However, drug d was included in the drug training set, $d \in D_{\text{train}}$, that is, there were interactions linking to d in the training set.

- E_4: Neither d nor p was included in the training set, $d \notin D_{\text{train}}$ and $p \notin P_{\text{train}}$, that is, there were no interactions linking to drug d or protein p in the training set.

In each strategy, we repeatedly performed 10 independent experiments with different random seeds when splitting each data set. The model with the best AUC on the validation set was selected and then assessed on the test set to compute the model performance. As the most widely used experimental setting, E_1 corresponded to evaluating the model performance in drug repositioning. Under E_1, we divided each data set into training, validation, and test sets with a ratio of 7:1:2. Settings E_2, E_3, and E_4 corresponded to more realistic situations on drug discovery and simulated different split situations of data sets. Under E_2, E_3, and E_4, 80% pairs were randomly selected for training and the remaining for test. Based on their different settings, drugs (or proteins) were removed, respectively. The remaining testing pairs were divided into the validation set and test set at a ratio of 1:2. Particularly, under the three settings, the training samples were sharply reduced.

3.2. Baseline Methods. We compared DO-GMA to the following six baselines.

CPI-GNN:[19] CPI-GNN uses an end-to-end representation learning framework for compound−protein interaction prediction by combining a GNN, convolutional neural network (CNN), and the neural attention mechanism.

BACPI:[43] BACPI is an end-to-end neural network model for predicting compound−protein interactions and binding affinity based on a graph attention network (GAT), CNN, and bidirectional attention neural network.

CPGL:[25] CPGL incorporates GAT and a long short-term memory neural network for DTI identification.

DrugBAN:[52] DrugBAN is a deep bilinear attention method with domain adaptation for inferring potential DTIs.

BINDTI:[49] BINDTI is a bidirectional intention framework that incorporates a GCN, ACmix, bidirectional intention network, and MLP.

FOTF-CPI:[37] FOTF-CPI is an optimal transport-based fragmentation model used to compute the binding affinity between compounds and proteins.

For the six DTI inference models described above, we followed the default hyperparameter settings provided in their original papers.

3.3. Comparison of Results. To evaluate the performances of our DO-GMA model under different realistic scenarios, we compared it with the other six competitors on four public data sets separately: DrugBank, BioSNAP, BindingDB, and C.elegans under four different experimental settings. The six benchmarks are CPI-GNN, BACPI, CPGL, DrugBAN, BINDTI, and FOTF-CPI. The results are shown in Tables 2−5, where the best results are marked in bold and the second-best results are marked as

Table 4. Performance Comparison under E_1 to E_4 on BindingDB (Running 10 Random Experiments)

method	AUC	AUPR	accuracy	F1-score	MCC
			Setting E_1		
CPI-GNN	0.5598 ± 0.0306	0.4700 ± 0.0289	0.5608 ± 0.0140	0.3362 ± 0.1691	0.0530 ± 0.0296
BACPI	0.9543 ± 0.0027	0.9415 ± 0.0038	0.8922 ± 0.0043	0.8710 ± 0.0058	0.7786 ± 0.0092
CPGL	0.9342 ± 0.0148	0.9140 ± 0.0204	0.8691 ± 0.0250	0.8435 ± 0.0282	0.7312 ± 0.0502
DrugBAN	0.9608 ± 0.0018	0.9473 ± 0.0033	0.9058 ± 0.0034	0.9073 ± 0.0026	0.8088 ± 0.0063
BINDTI	0.9609 ± 0.0017	0.9465 ± 0.0027	**0.9073 ± 0.0034**	**0.9089 ± 0.0027**	**0.8118 ± 0.0065**
FOTF-CPI	0.9531 ± 0.0023	0.9370 ± 0.0036	0.8936 ± 0.0034	0.8730 ± 0.0038	0.7816 ± 0.0068
DO-GMA	**0.9624 ± 0.0024**	**0.9501 ± 0.0029**	0.9012 ± 0.0046	0.9051 ± 0.0041	0.8010 ± 0.0087
			Setting E_2		
CPI-GNN	0.5098 ± 0.0293	0.4292 ± 0.0265	0.5784 ± 0.0063	0.0483 ± 0.1450	0.0114 ± 0.0341
BACPI	0.9317 ± 0.0039	0.9093 ± 0.0057	0.8619 ± 0.0040	0.8404 ± 0.0050	0.7202 ± 0.0082
CPGL	0.9106 ± 0.0167	0.8807 ± 0.0202	0.8356 ± 0.0231	0.8097 ± 0.0261	0.6664 ± 0.0462
DrugBAN	0.9388 ± 0.0026	0.9142 ± 0.0057	0.8738 ± 0.0067	0.8776 ± 0.0036	0.7459 ± 0.0112
BINDTI	**0.9402 ± 0.0043**	0.9137 ± 0.0078	**0.8741 ± 0.0064**	**0.8799 ± 0.0051**	**0.7475 ± 0.0119**
FOTF-CPI	0.9274 ± 0.0038	0.8989 ± 0.0046	0.8569 ± 0.0051	0.8337 ± 0.0058	0.7091 ± 0.0099
DO-GMA	0.9390 ± 0.0043	**0.9148 ± 0.0065**	0.8670 ± 0.0072	0.8735 ± 0.0057	0.7339 ± 0.0129
			Setting E_3		
CPI-GNN	0.5299 ± 0.0306	0.4308 ± 0.0243	0.5281 ± 0.0205	0.4541 ± 0.0371	0.0408 ± 0.0400
BACPI	0.5346 ± 0.0162	0.4431 ± 0.0238	0.5371 ± 0.0169	0.4346 ± 0.0557	0.0457 ± 0.0205
CPGL	0.5773 ± 0.0218	0.4906 ± 0.0216	**0.5406 ± 0.0286**	0.5248 ± 0.0258	0.1043 ± 0.0320
DrugBAN	0.5911 ± 0.0133	0.5183 ± 0.0188	0.4208 ± 0.0104	0.6670 ± 0.0003	0.0314 ± 0.0247
BINDTI	0.5589 ± 0.0144	0.4695 ± 0.0173	0.4210 ± 0.0117	0.6671 ± 0.0004	0.0333 ± 0.0245
FOTF-CPI	0.5648 ± 0.0105	0.4907 ± 0.0190	0.5354 ± 0.0356	0.4837 ± 0.0630	0.0780 ± 0.0219
DO-GMA	**0.6279 ± 0.0136**	**0.5351 ± 0.0185**	0.4671 ± 0.0300	**0.6731 ± 0.0031**	**0.1254 ± 0.0388**
			Setting E_4		
CPI-GNN	0.4987 ± 0.0149	0.4185 ± 0.0140	0.5130 ± 0.0131	0.4196 ± 0.0438	0.0009 ± 0.0243
BACPI	0.5050 ± 0.0144	0.4251 ± 0.0130	0.5212 ± 0.0338	0.3756 ± 0.1516	0.0075 ± 0.0223
CPGL	0.5453 ± 0.0280	0.4721 ± 0.0392	**0.5535 ± 0.0270**	0.3637 ± 0.1438	**0.0549 ± 0.0435**
DrugBAN	0.5475 ± 0.0250	0.4710 ± 0.0262	0.4208 ± 0.0057	0.6668 ± 0.0001	0.0167 ± 0.0077
BINDTI	0.5267 ± 0.0140	0.4398 ± 0.0101	0.4212 ± 0.0052	0.6669 ± 0.0001	0.0199 ± 0.0087
FOTF-CPI	0.5367 ± 0.0244	0.4661 ± 0.0272	0.4840 ± 0.0381	0.5398 ± 0.0386	0.0440 ± 0.0309
DO-GMA	**0.5603 ± 0.0241**	**0.4803 ± 0.0281**	0.4262 ± 0.0076	**0.6677 ± 0.0014**	0.0340 ± 0.0319

underlined for each evaluation metric. Figures 6 and 7 depict their ROC and PR curves. Particularly, all comparison experiments were performed under 10 random shuffles on each data set.

The results in Tables 2−5 show that, on the DrugBank and C.elegans data sets, DO-GMA outperformed the six baselines according to the AUC, AUPR, accuracy, F1-score, and MCC under E_1, E_2, E_3, and E_4. In particular, on DrugBank, compared to the best-performing baseline, under E_1, DO-GMA surpassed it in these categories by 2.92%, 3.47%, 2.11%, 2.35%, and 4.27%, respectively. Under E_2, it outperformed by 3.86%, 3.74%, 2.71%, 2.32%, and 4.04%, respectively. Under E_3, it improved by 8.56%, 5.36%, 1.80%, 2.04%, and 5.14%. Under E_4, the corresponding improvements were 4.23%, 2.68%, 2.44%, 6.36%, and 8.65%, respectively.

On BioSNAP, the performance of DO-GMA over the other six competitors was improved under the four settings in the vast majority of cases. On BindingDB, DO-GMA computed the highest AUC and AUPR among all seven DTI prediction methods under all four experimental settings in most cases. Moreover, on BindingDB, although DO-GMA slightly lags behind DrugBAN and BINDTI based on accuracy, F1-score, and MCC, the performance difference between it and the two baselines based on accuracy, F1-score, and MCC was very small.

Moreover, CPI-GNN, BACPI, DrugBAN, BINDTI, and FOTF-CPI used an attention mechanism for DTI prediction. However, our DO-GMA framework improved the conventional attention mechanism and proposed a gated multihead attention mechanism. On all four data sets, DO-GMA exceeded the six state-of-the-art baselines in most cases, demonstrating the effectiveness of the gated multihead attention.

Additionally, under E_1, DO-GMA and the six baselines computed the optimal performance on all four data sets. Under E_2, E_3, and E_4, their performance decreased compared with that under E_1, which may be caused by fewer training samples. E_4 had the least training samples; thus, all seven methods calculated the lowest performance.

3.4. Ablation Experiments. In our DO-GMA framework, we combined DO-Conv, GCN, the gated mechanism, and the multihead attention mechanism with shared-learned queries and bilinear model concatenation for DTI prediction. To explore how different modules influence predictions, we conducted multiple ablation experiments on four data sets:

- With DO-Conv or not: To validate the effectiveness of DO-Conv on drug and protein representation learning, we replaced DO-Conv with a regular CNN to extract drug features from their SMILES strings and protein features from their amino acid sequences.
- Unimodality or multimodality: To evaluate the role of drug unimodality feature learning and multimodality feature learning on predictions, we analyzed two DO-GMA variants: Uni-S (drug SMILES sequence + protein sequence) and Uni-M (drug 2D molecular graph + protein sequence).

Table 5. Performance Comparison under E_1 to E_4 on C.elegans (Running 10 Random Experiments)

method	AUC	AUPR	accuracy	F1-score	MCC
			Setting E_1		
CPI-GNN	0.9536 ± 0.0079	0.9422 ± 0.0121	0.9146 ± 0.0100	0.9185 ± 0.0088	0.8292 ± 0.0201
BACPI	0.9864 ± 0.0044	0.9869 ± 0.0039	0.9489 ± 0.0114	0.9487 ± 0.0117	0.8981 ± 0.0227
CPGL	0.9757 ± 0.0034	0.9797 ± 0.0029	0.9265 ± 0.0117	0.9288 ± 0.0103	0.8538 ± 0.0224
DrugBAN	0.9865 ± 0.0044	0.9856 ± 0.0040	0.9619 ± 0.0071	0.9619 ± 0.0072	0.9240 ± 0.0140
BINDTI	0.9893 ± 0.0022	0.9892 ± 0.0029	0.9662 ± 0.0061	0.9662 ± 0.0061	0.9325 ± 0.0122
FOTF-CPI	0.9919 ± 0.0030	0.9909 ± 0.0054	0.9663 ± 0.0051	0.9661 ± 0.0052	0.9328 ± 0.0101
DO-GMA	**0.9933 ± 0.0027**	**0.9934 ± 0.0028**	**0.9740 ± 0.0057**	**0.9738 ± 0.0058**	**0.9480 ± 0.0114**
			Setting E_2		
CPI-GNN	0.6830 ± 0.0649	0.7278 ± 0.0391	0.6320 ± 0.0584	0.6038 ± 0.0763	0.2844 ± 0.1126
BACPI	0.8083 ± 0.0280	0.8315 ± 0.0294	0.7079 ± 0.0272	0.6653 ± 0.0427	0.4435 ± 0.0526
CPGL	0.8625 ± 0.0231	0.8897 ± 0.0130	0.7546 ± 0.0323	0.7335 ± 0.0587	0.5464 ± 0.0500
DrugBAN	0.8366 ± 0.0181	0.8773 ± 0.0132	0.7901 ± 0.0151	0.7753 ± 0.0173	0.5921 ± 0.0344
BINDTI	0.8385 ± 0.0258	0.8746 ± 0.0175	0.7686 ± 0.0209	0.7678 ± 0.0195	0.5440 ± 0.0462
FOTF-CPI	0.8712 ± 0.0213	0.8913 ± 0.0188	0.7861 ± 0.0228	0.7605 ± 0.0415	0.5993 ± 0.0334
DO-GMA	**0.8718 ± 0.0216**	**0.8954 ± 0.0140**	**0.8027 ± 0.0131**	**0.7969 ± 0.0191**	**0.6101 ± 0.0226**
			Setting E_3		
CPI-GNN	0.8998 ± 0.0330	0.9007 ± 0.0419	0.8351 ± 0.0409	0.8199 ± 0.0531	0.6797 ± 0.0810
BACPI	0.9438 ± 0.0166	0.9528 ± 0.0106	0.8657 ± 0.0183	0.8422 ± 0.0232	0.7494 ± 0.0292
CPGL	0.8641 ± 0.0231	0.8868 ± 0.0234	0.7644 ± 0.0456	0.7133 ± 0.0743	0.5666 ± 0.0759
DrugBAN	0.9475 ± 0.0095	0.9491 ± 0.0086	0.8832 ± 0.0193	0.8858 ± 0.0151	0.7696 ± 0.0352
BINDTI	0.9512 ± 0.0064	0.9560 ± 0.0070	0.8995 ± 0.0122	0.8967 ± 0.0122	0.7993 ± 0.0241
FOTF-CPI	0.9438 ± 0.0096	0.9514 ± 0.0082	0.8497 ± 0.0211	0.8181 ± 0.0306	0.7238 ± 0.0349
DO-GMA	**0.9580 ± 0.0122**	**0.9573 ± 0.0147**	**0.9041 ± 0.0140**	**0.9012 ± 0.0151**	**0.8085 ± 0.0277**
			Setting E_4		
CPI-GNN	0.5842 ± 0.0588	0.5717 ± 0.0515	0.5166 ± 0.0423	0.3096 ± 0.1012	0.0625 ± 0.1052
BACPI	0.6662 ± 0.0405	0.6583 ± 0.0342	0.5621 ± 0.0333	0.2974 ± 0.1069	0.1715 ± 0.0724
CPGL	0.6827 ± 0.0608	0.6942 ± 0.0759	0.5028 ± 0.0389	0.0852 ± 0.1722	0.0385 ± 0.0873
DrugBAN	0.6490 ± 0.0476	0.6534 ± 0.0569	0.5630 ± 0.0484	0.6848 ± 0.0154	0.1907 ± 0.0870
BINDTI	0.6932 ± 0.0464	0.7039 ± 0.0341	0.5866 ± 0.0602	0.6939 ± 0.0239	0.2374 ± 0.1036
FOTF-CPI	0.6561 ± 0.0355	0.6430 ± 0.0423	0.5577 ± 0.0254	0.2914 ± 0.1338	0.1653 ± 0.0635
DO-GMA	**0.7036 ± 0.0376**	**0.7062 ± 0.0379**	**0.6072 ± 0.0607**	**0.6958 ± 0.0190**	**0.2571 ± 0.1085**

- With the gated attention mechanism or not: To confirm the effect of the gated attention mechanism on the feature enhancement of drugs and targets, we removed it from DO-GMA.
- With the multihead attention mechanism or not: To test the role of the multihead attention mechanism on feature fusion, we removed it from DO-GMA.

Figure 8 gives the performance of ablation experiments under all four settings. From the results in Figure 8, we found that (1) DO-Conv, multimodality representation learning, the gated attention, and the multihead attention improved predictions, especially the multihead attention, followed by the gated attention. (2) The full model leveraged the strengths of four modules and outperformed them.

3.5. Robust Statistical Analysis. The Wilcoxon signed-rank test can reveal significant difference between two different methods.[64] To evaluate the robustness of our model, we used the Wilcoxon signed-rank test for statistical analysis using the SPSS software based on the AUCs and AUPRs computed by DO-GMA and the six baselines. The results are reported in Tables 6−9.

As shown in Tables 6−9, on DrugBank, BioSNAP, and BindingDB, the p-values between DO-GMA and the six baselines were far less than 0.05, indicating that DO-GMA obtained a more significant improvement compared to the six baselines. On C.elegans, the p-values between DO-GMA and FOTF-CPI based on their computed AUCs and AUPRs were 0.059336 and 0.139414, respectively. It seems the difference between FOTF-CPI and DO-GMA was not significant, which may be caused by the small size of the data set. However, the p-values between DO-GMA and CPI-GNN, BACPI, CPGL, DrugBAN, and BINDTI were obviously less than 0.05, again demonstrating that the DO-GMA performance has been significantly improved. In general, DO-GMA demonstrated better robustness compared to the six baselines.

3.6. Sensitivity Analysis of Parameters. To evaluate the effects of parameters on DTI prediction, we discussed the performance of DO-GMA under different settings of three important hyperparameters, i.e., epoch size, learning rate, and number of GCN layers.

We evaluated the DO-GMA performance under different epoch sizes with the range of [1, 200] on the four DTI data sets. As shown in Figure 9, when the epoch size increased from 1 to 100, the performance of DO-GMA was gradually boosted based on its computed AUC, AUPR, F1-score, and MCC. When the epoch size was around 100, it tended to be relatively stable. When the epoch size continued to increase, its improvement was very little. Therefore, we set the epoch size to 100.

We still analyzed the DO-GMA performance under different learning rates on the four data sets, i.e., 1e-4, 5e-4, 5e-5, 1e-5, and 5e-6. As shown in Figure 10, when the learning rate decreased from 1e-4 to 5e-4, the performance of DO-GMA decreased based on its computed AUC, AUPR, F1-score, and MCC. When the learning rate changed from 5e-4 to 5e-5, its performance was

https://doi.org/10.1021/acs.jcim.4c02088
J. Chem. Inf. Model. 2025, 65, 1318−1337

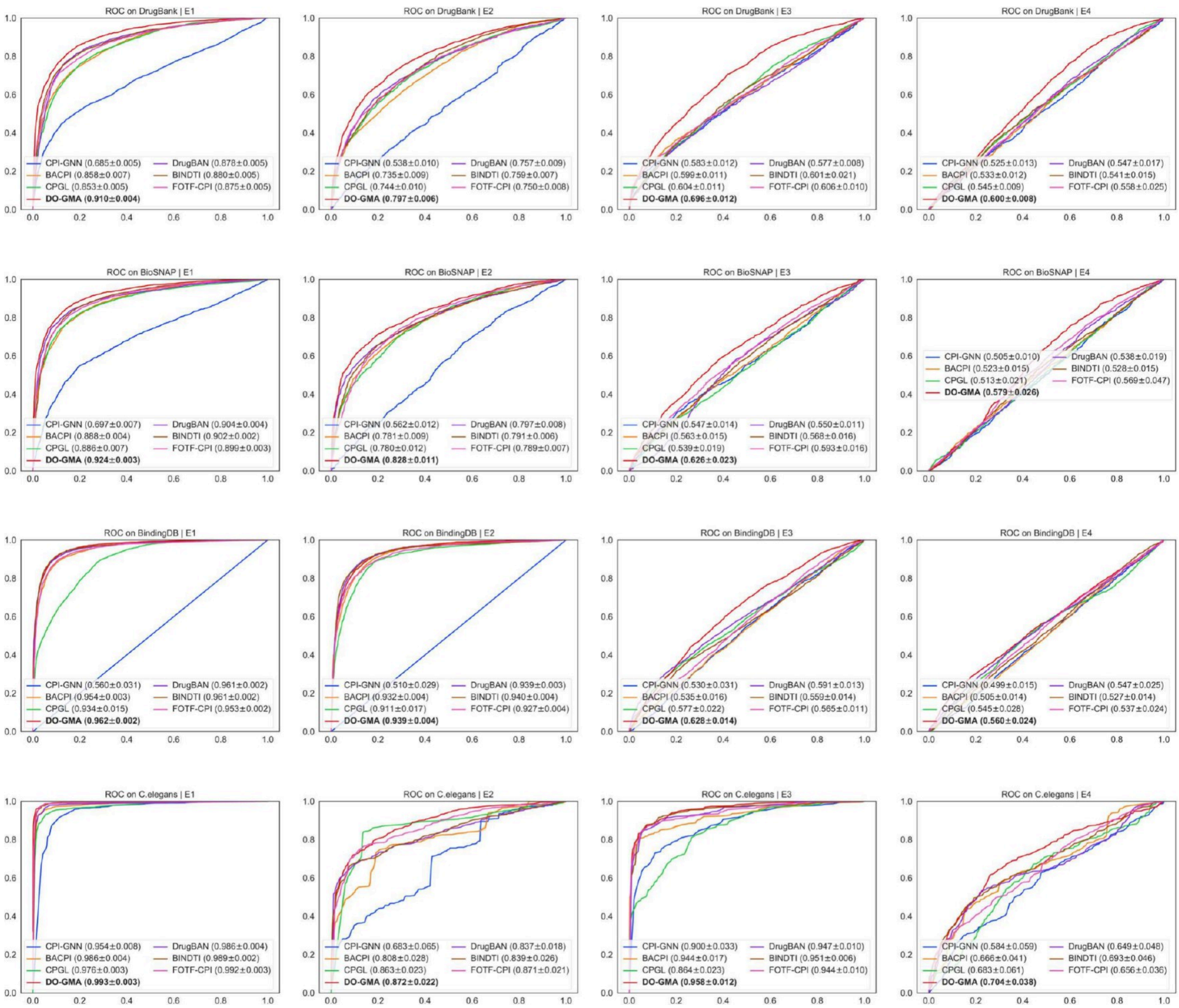


Figure 6. ROC curves of DO-GMA and six baselines under four experimental settings on four data sets.

promoted. When the learning rate was reduced from 5e-5 to 1e-5 and from 1e-5 to 5e-6, the performance reduced. Therefore, we set the learning rate to 5e-5.

Finally, we investigated the DO-GMA performance under different numbers of GCN layers on the four data sets. As shown in Figure 11, when the number of GCN layers increased from 1 to 3, the DO-GMA performance increased based on its computed AUC, AUPR, F1-score, and MCC. When the layer number changed from 3 to 5, its performance gradually decreased. Therefore, we set the number of GCN layers to 3.

3.7. Visualization. Our DO-GMA model fused drug features based on their SMILES strings and 2D molecular graphs and protein features based on their amino acid sequences. To characterize the distribution of the above features, we performed dimensionality reduction on the fused features and visualized them using UMAP. In particular, each data set was visualized based on three types of data:

- V1: Drug SMILES string features extracted by DO-Conv, drug 2D molecular graph features extracted by GCN, and protein sequence features extracted by DO-Conv. The three features were enhanced by the gated attention mechanism.
- V2: The fused features of DTPs with real labels.
- V3: The fused features of DTPs with the predicted labels.

The visualization results on the four DTI data sets are shown in Figure 12. We found that DO-GMA can effectively map drug and protein features to different regions, thereby generating denser drug and protein subpopulations. Furthermore, the visualized results on the fused features with the predicted labels were basically consistent with those with true labels, elucidating its powerful DT feature learning ability for downstream prediction.

3.8. Computational Cost Analysis. Using multiply-accumulate operations (MACs),[65] which are used to evaluate the computational complexity of a model, the number of model parameters, and GPU memory consumption, we further analyzed the computational cost of DO-GMA and the other six baselines. Smaller MACs, the number of model parameters, and GPU memory consumption denote less computational cost of the model. As shown in Table 10, our proposed DO-GMA model had smaller MACs than DrugBAN, BINDTI, and FOTF-

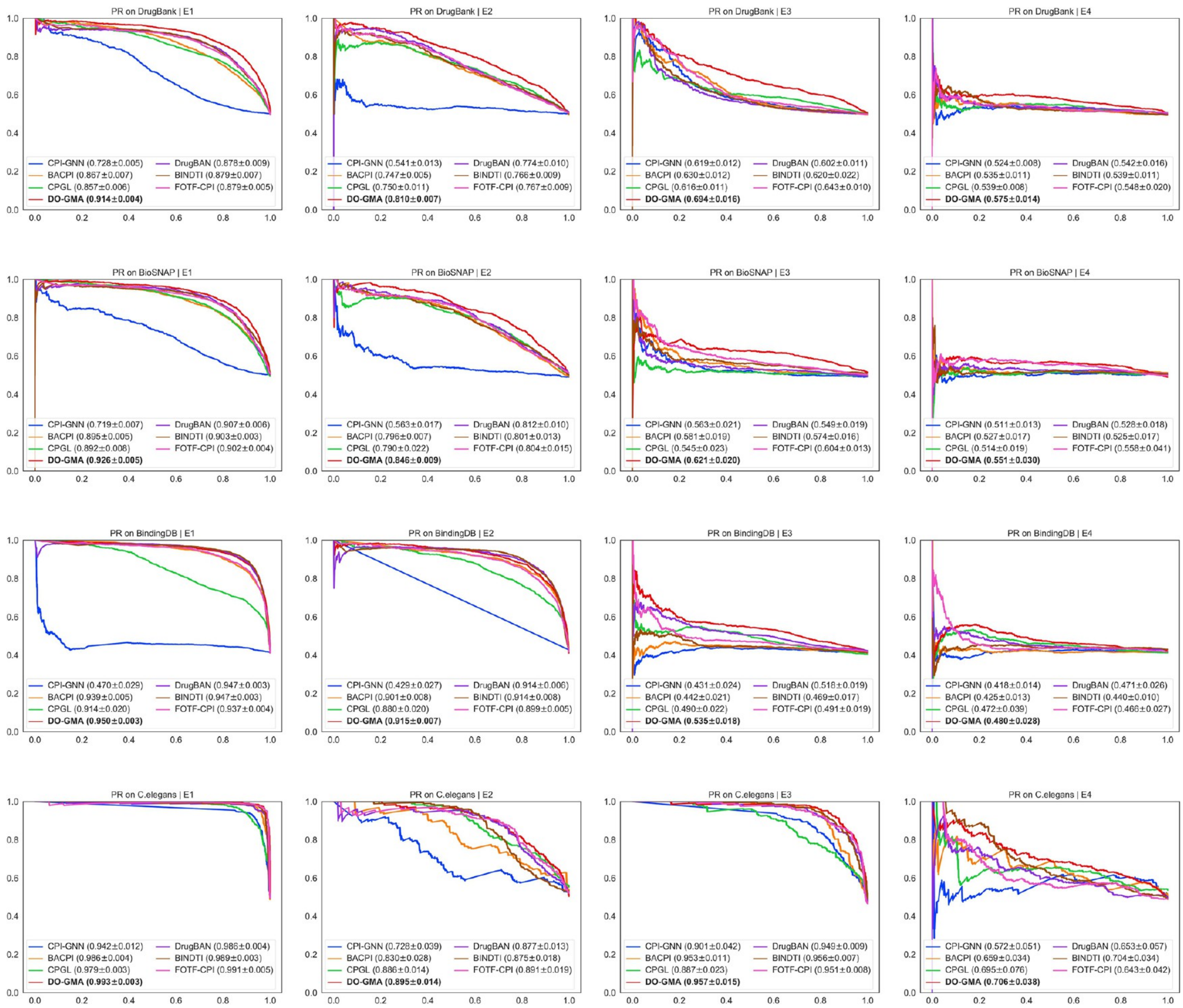


Figure 7. PR curves of DO-GMA and six baselines under four experimental settings on four data sets.

CPI. The number of total parameters used by DO-GMA was 0.753M, which was less than that in DrugBAN, BINDTI, and FOTF-CPI. It had reduced GPU memory consumption compared to BACPI, CPGL, DrugBAN, BINDTI, and FOTF-CPI. Although DO-GMA needs more MACs, a greater number of parameters, and more GPU memory than CPI-GNN, along with requiring more MACs and a larger number of parameters than BACPI and CPGL, it significantly surpasses the three models based on AUC, AUPR, F1-score, and MCC. DrugBAN, BINDTI, and FOTF-CPI are three recent state-of-the-art deep learning-based DTI prediction models, but DO-GMA has less computational cost while computing a better performance than these three methods, further demonstrating its powerful DTI inference ability.

3.9. Case Study. In this section, we performed case studies from drug and target spaces to further validate the reliability of our DO-GMA model. In case analyses, all interactions involving the query drug (or target) were removed, and interaction probabilities between the drug (or target) and other targets (or drugs) were computed. Finally, we ranked these drug–target pairs according to their probabilities.

For the drug space, cinnarizine (DB00568) and primidone (DB00794) were selected to perform case studies. Cinnarizine is an antihistamine and calcium channel blocker.[66] It can increase cerebral blood flow for patients with cerebral atherosclerosis, stroke, and post-traumatic brain symptoms.[67−69] Primidone is mainly applied to complex partial and generalized tonic-clonic seizures.[70,71] It is the most efficacious drug used for the treatment of essential tremor.[72] As shown in Table 11, among the predicted top 10 targets interacting with cinnarizine and primidone, almost all of the top predictions are included in the DrugBank database. In addition, DO-GMA predicted that P06276 could interact with DB00568.

For the target space, D(4) dopamine receptor (P21917) and sodium channel protein type 8 subunit alpha (Q9UQD0) were used to perform case studies. P21917 plays a regulation role in calcium-mediated signaling, circadian rhythm, dopamine metabolism, and neurotransmitter secretion.[73] It can mediate the effects of atypical antipsychotics.[74] Q9UQD0 has a dense correlation with ATP binding and voltage-gated sodium channel activity. It plays an important role in nervous system development, muscle organ development, myelin formation,

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J. Chem. Inf. Model. 2025, 65, 1318−1337

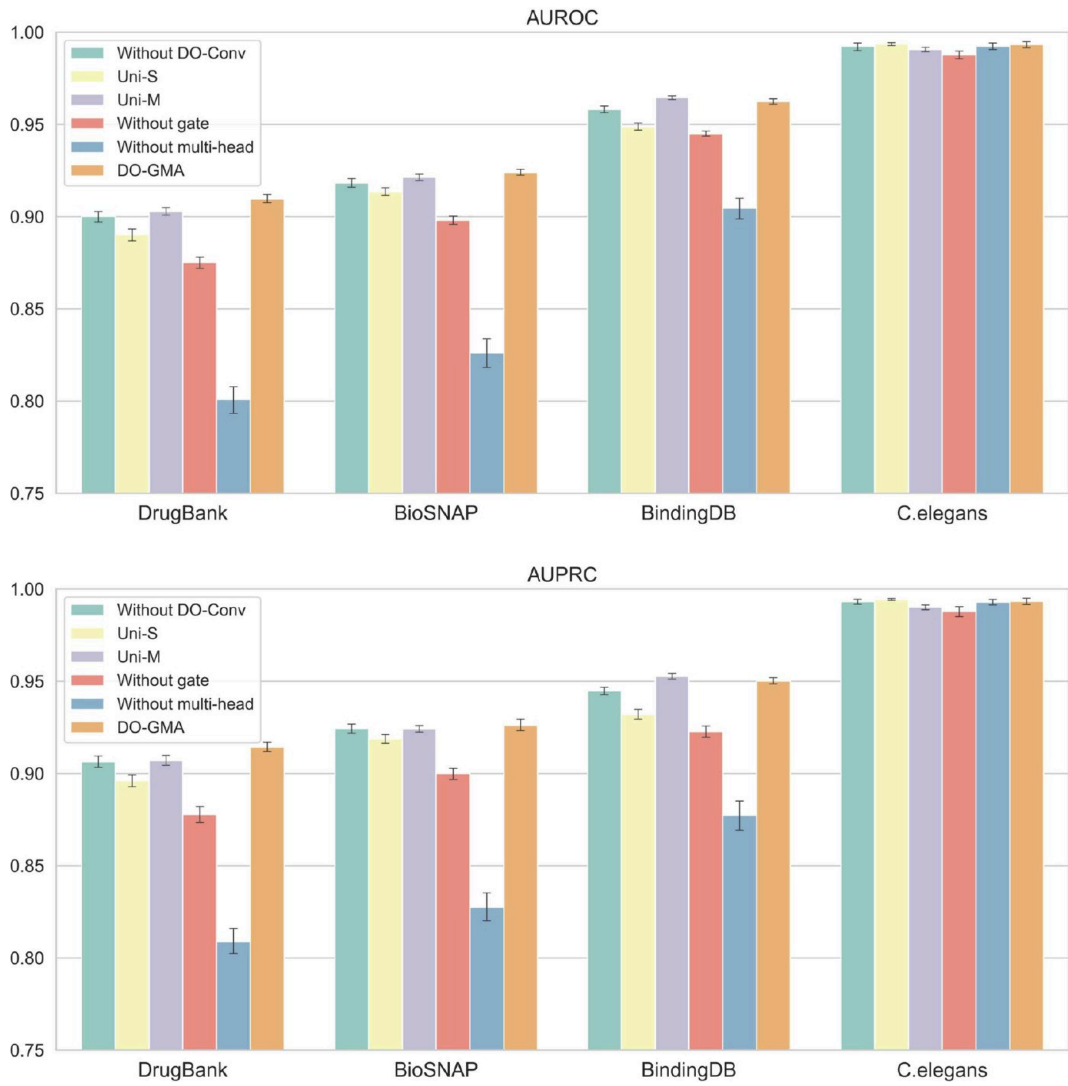

Figure 8. Ablation experiments (running 10 random experiments).

Table 6. Robust Statistical Analysis of DO-GMA and Six Baselines on DrugBank

		p-value	
ours	others	AUC	AUPR
DO-GMA	CPI-GNN	0.005062	0.005062
	BACPI	0.005062	0.005062
	CPGL	0.005062	0.005062
	DrugBAN	0.005062	0.005062
	BINDTI	0.005062	0.005062
	FOTF-CPI	0.005062	0.005062

Table 7. Robust Statistical Analysis of DO-GMA and Six Baselines on BioSNAP

		p-value	
ours	others	AUC	AUPR
DO-GMA	CPI-GNN	0.005062	0.005062
	BACPI	0.005062	0.005062
	CPGL	0.005062	0.005062
	DrugBAN	0.005062	0.005062
	BINDTI	0.005062	0.005062
	FOTF-CPI	0.005062	0.005062

Table 8. Robust Statistical Analysis of DO-GMA and Six Baselines on BindingDB

		p-value	
ours	others	AUC	AUPR
DO-GMA	CPI-GNN	0.005062	0.005062
	BACPI	0.005062	0.005062
	CPGL	0.005062	0.005062
	DrugBAN	0.046853	0.036658
	BINDTI	0.028417	0.036658
	FOTF-CPI	0.005062	0.005062

Table 9. Robust Statistical Analysis of DO-GMA and Six Baselines on C.elegans

		p-value	
ours	others	AUC	AUPR
DO-GMA	CPI-GNN	0.005062	0.005062
	BACPI	0.005062	0.005062
	CPGL	0.005062	0.005062
	DrugBAN	0.005062	0.005062
	BINDTI	0.006910	0.009344
	FOTF-CPI	0.059336	0.139414

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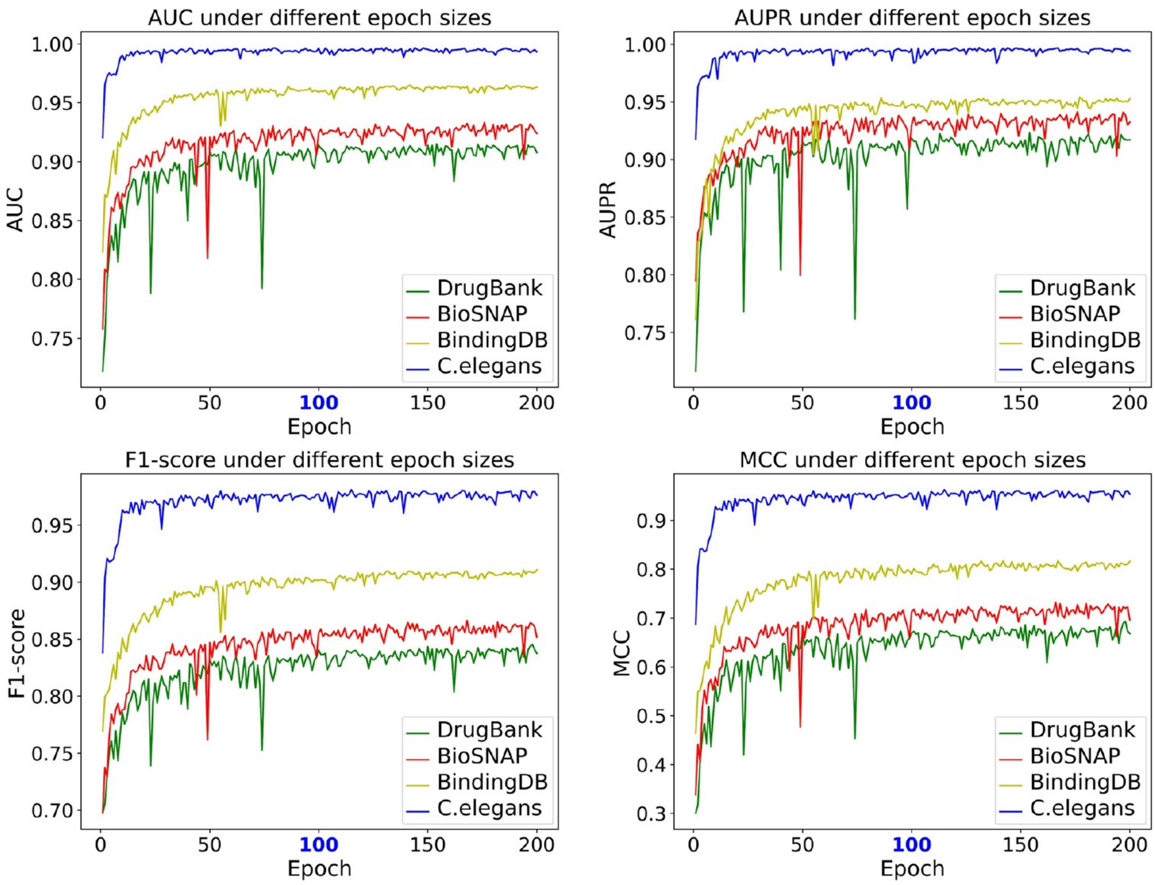

Figure 9. Performance of DO-GMA under different epoch sizes.

and adult walking behavior.[73] As shown in Table 12, among the inferred top 10 drugs targeting P21917 and Q9UQD0, only one interaction (i.e., interaction between Q9UQD0 and DB09118) has not been verified.

To validate the predicted interactions (i.e., DB00568 and P06276, and DB09118 and Q9UQD0) during the case studies, we conducted molecular docking for them using AutoDock4.[75] As shown in Figures 13 and 14, their docking results were visualized by PyMOL.[76] Particularly, they had high molecular binding energies of −5.92 and −5.69 kcal/mol, respectively, demonstrating their strong binding stabilities. The above results further validated the strong adaptability of DO-GMA in DTP classification tasks.

4. DISCUSSION

DTI prediction is an important task in drug discovery. Recent development in large-scale experimental techniques has greatly promoted drug discovery and target identification.[77] However, wet experiments for DTI prediction is a labor-intensive, time-consuming, and complex process. Compared to wet lab experiments, computational methods allow high-throughout predictions by learning intrinsic features of drugs and targets and inferring their potential interactions simultaneously.[78] Therefore, researchers have been increasingly focused on developing computational methods to find a list of promising DTI candidates while greatly reducing the enormous search scope of DTPs, thereby guiding drug design and improving drug efficacy.

Recently, deep learning methods, including graph-based, transformer-based, and attention-based methods, have provided good indicators for identifying potential DTIs. CPI-GNN and CPGL are graph-based DTI prediction models. CPI-GNN learned protein features through word embedding, which can extract more complex context information than the amino acid-based embedding used in DeepDTA,[79] and improved DTI prediction to some extent. CPGL regarded compound molecules as graphs and learned compound representations through GAT, and extracted protein features via a long short-term memory network, thereby inferring potential DTIs in an end-to-end manner. FOTF-CPI is a transformer-based DTI prediction model. It fused an optimal transport-based fragmentation model and attention mechanisms for DTP classification. BACPI, DrugBAN, and BINDTI are attention-based methods. BACPI utilized GNN and CNN to extract DTP features and enhanced these features through an attention mechanism. DrugBAN explicitly captured local pairwise DTPs through a deep bilinear attention network with domain adaptation. BINDTI improved DTI prediction using a bidirectional intention network.

The aforementioned deep learning models effectively shed light on the potential interactions. However, the performance of CPI-GNN, CPGL, FOTF-CPI, and BACPI needs further improvement, and DrugBAN and BINDTI require high computational resources. Moreover, they learn drug feature representations from only a single perspective, which is insufficient to comprehensively characterize drug biological features. Additionally, feature fusion routes for drugs and proteins need further exploration. Here, we developed a novel end-to-end framework named DO-GMA for potential DTI identification based on DO-Conv and the gated multihead

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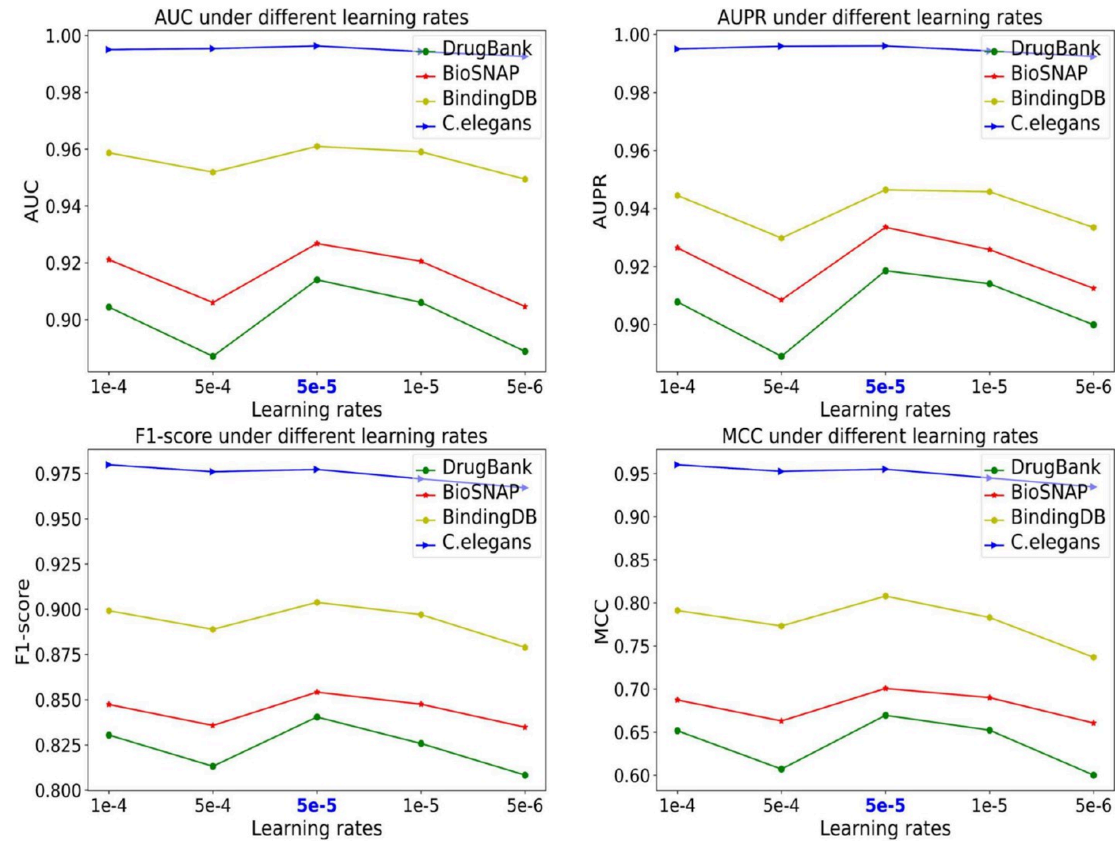

Figure 10. Performance of DO-GMA under different learning rates.

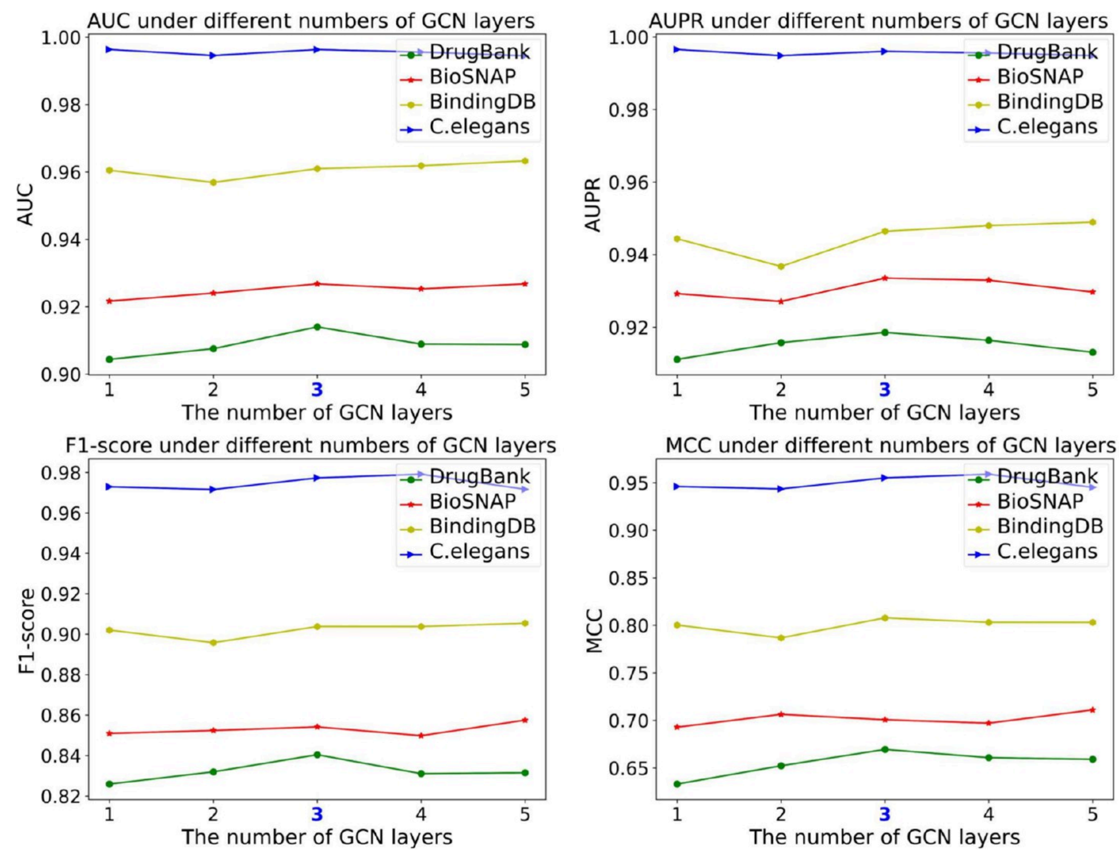

Figure 11. Performance of DO-GMA under different numbers of GCN layers.

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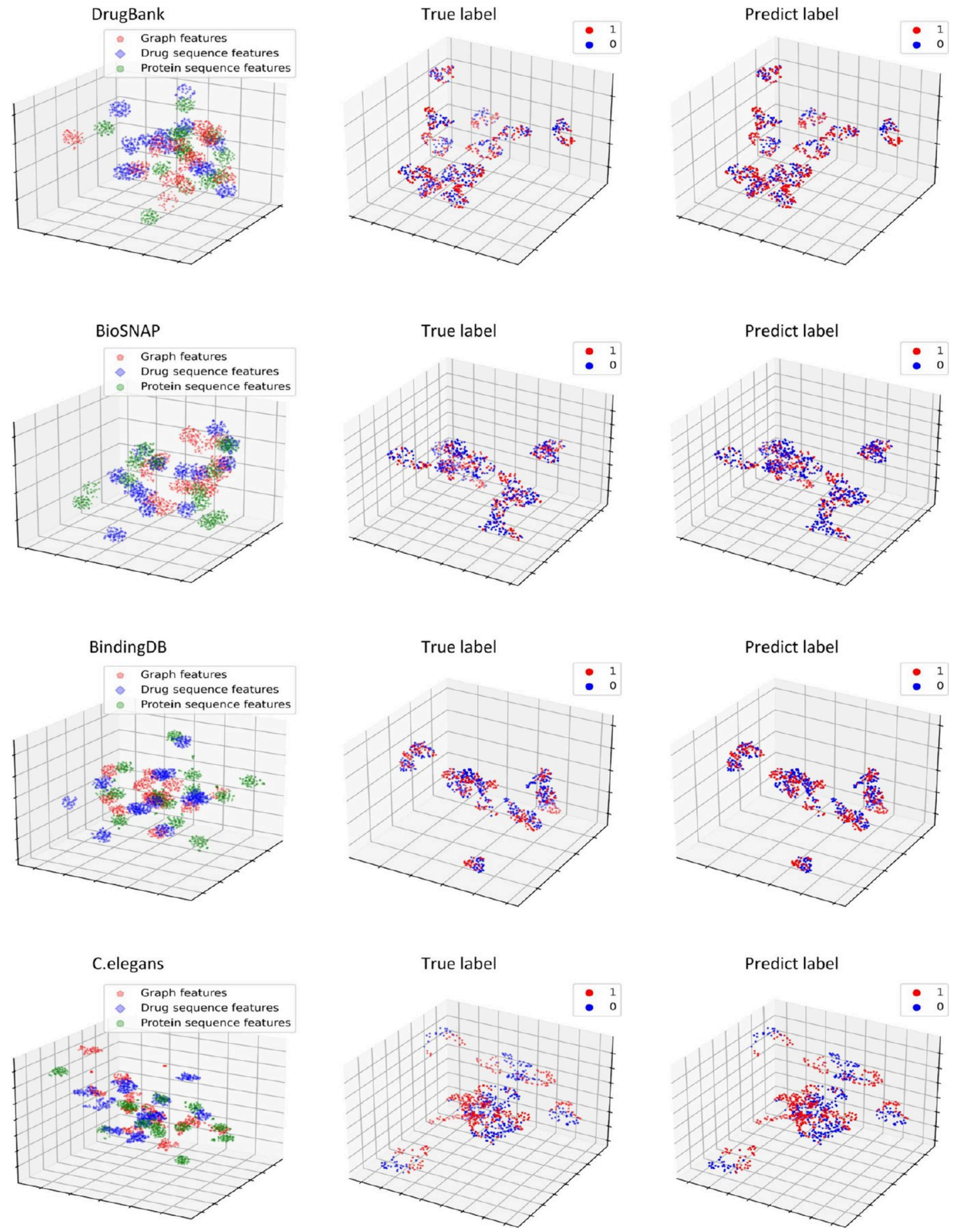

Figure 12. Visualization of the fused features on four data sets. The three subfigures from left to right correspond to V1, V2, and V3, respectively. 0 and 1 represent DT pairs without interaction and with interaction, respectively.

Table 10. Computational Cost of DO-GMA and Six Baselines Based on MACs, the Number of Parameters, and GPU Memory Consumption

method	MACs	params	GPU memory
CPI-GNN	0.397G	0.003M	1765MB
BACPI	2.821G	0.195M	24659MB
CPGL	0.170G	0.012M	5237MB
DrugBAN	33.003G	1.017M	3401MB
BINDTI	21.926G	0.875M	4235MB
FOTF-CPI	135.493G	36.128M	16539MB
DO-GMA	11.536G	0.753M	2435MB

attention mechanism with shared-learned queries and bilinear model concatenation.

DO-GMA first learned drug representations from their SMILES strings and protein features from their amino acid sequences using DO-Conv. Next, it extracted drug representations from their 2D molecular graphs through GCN. Subsequently, it incorporated the gated attention mechanism and the multihead attention mechanism with shared-learned queries and bilinear model concatenation to fuse drug and protein features. Finally, it took the fused DTP features as inputs and built an MLP to classify unlabeled DTPs.

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J. Chem. Inf. Model. 2025, 65, 1318−1337

Table 11. Predicted Top 10 Targets Interacting with Cinnarizine (DB00568) and Primidone (DB00794)

	DB00568		DB00794	
rank	UniProt ID	result	UniProt ID	result
1	P21728	TRUE	P14867	TRUE
2	P35367	TRUE	P36544	TRUE
3	P08173	TRUE	P47869	TRUE
4	P14416	TRUE	Q9UN88	TRUE
5	P08912	TRUE	O14764	TRUE
6	P20309	TRUE	Q16445	TRUE
7	P11229	TRUE	P34903	TRUE
8	P08172	TRUE	P18505	TRUE
9	P06276	Unknown	P31644	TRUE
10	P21918	TRUE	P18507	TRUE

Table 12. Predicted Top 10 Drugs Targeting P21917 and Q9UQD0

	P21917		Q9UQD0	
rank	DrugBank ID	result	DrugBank ID	result
1	DB01186	TRUE	DB00243	TRUE
2	DB05271	TRUE	DB09342	TRUE
3	DB00248	TRUE	DB00273	TRUE
4	DB00589	TRUE	DB11186	TRUE
5	DB13345	TRUE	DB09085	TRUE
6	DB09286	TRUE	DB00564	TRUE
7	DB11275	TRUE	DB05541	TRUE
8	DB01403	TRUE	DB09118	Unknown
9	DB00777	TRUE	DB09088	TRUE
10	DB01200	TRUE	DB13269	TRUE

To validate the performance of DO-GMA, we performed a comprehensive comparative experiment. First, DO-GMA was benchmarked against six baseline DTI prediction models (CPI-GNN, BACPI, CPGL, DrugBAN, BINDTI, and FOTF-CPI) using four data sets (i.e., DrugBank, BioSNAP, C.elegans, and BindingDB) under four different experimental settings (E_1, E_2, E_3, and E_4). Through evaluation with AUC, AUPR, accuracy, F1-score, and MCC, DO-GMA obviously surpassed the six baselines above, elucidating its greater robustness and generalization ability. Moreover, compared to five attention mechanism-based methods (i.e., CPI-GNN, BACPI, DrugBAN, BINDTI, and FOTF-CPI), DO-GMA computed the best performance, validating that the gated multihead attention mechanism facilitates acquiring potential interaction representations between drugs and targets.

Moreover, we performed ablation experiments. The results demonstrated that the combination of DO-Conv and GCN could effectively extract drug and protein features, demonstrating that drug multimodal representations facilitate the improvement of DTI prediction. Moreover, the gated attention mechanism could enhance DTP feature learning, and the multihead attention mechanism could better fuse drug and protein features by combining shared-learned queries and bilinear model concatenation.

Furthermore, drug multimodal features and protein sequence features were visualized. The visualization results precisely characterize the distributions of three different types of features. Finally, we conducted case studies from drug and target spaces. The results from database retrieval, molecular docking, and visualization showed that there could be interactions between P06276 and DB00568 as well as between Q9UQD0 and DB09118.

DO-GMA can accurately identify DTI candidates, which may be attributed to the following three features. (i) GMA combined the gated attention and the multihead attention with shared-learned queries and bilinear model concatenation. It better fused drug and protein features and improved the interpretability and generalization ability of DO-GMA. (ii) Most traditional drug representation learning methods are based on the unimodal strategy and fail to fully capture drug abundant features. In contrast, we learned drug multimodal features by the combination of DO-Conv and GCN and thus acquired more detailed feature information. (iii) Most current drug representation learning methods used traditional convolution, but we used DO-Conv for drug feature extraction. Our DO-GMA model converged faster and performed more accurately.

Compared with other state-of-the-art methods, DO-GMA achieved a superior performance, but it can still be improved. Proteins demonstrate abundant 3D structure information. This information significantly facilitates protein feature extraction. Hence, in the future, we will fully combine their 3D structures for promoting DTI prediction through popular tools like DeepMind's AlphaFold.[80]

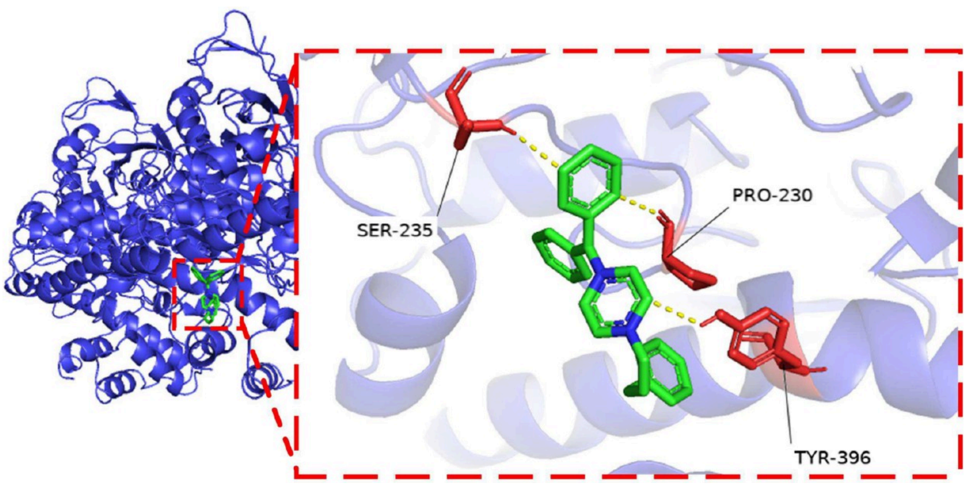

Figure 13. Visualization of molecular docking for DB00568 and P06276. Drug: DB00568, protein: P06276, affinity: −5.92 (kcal/mol).

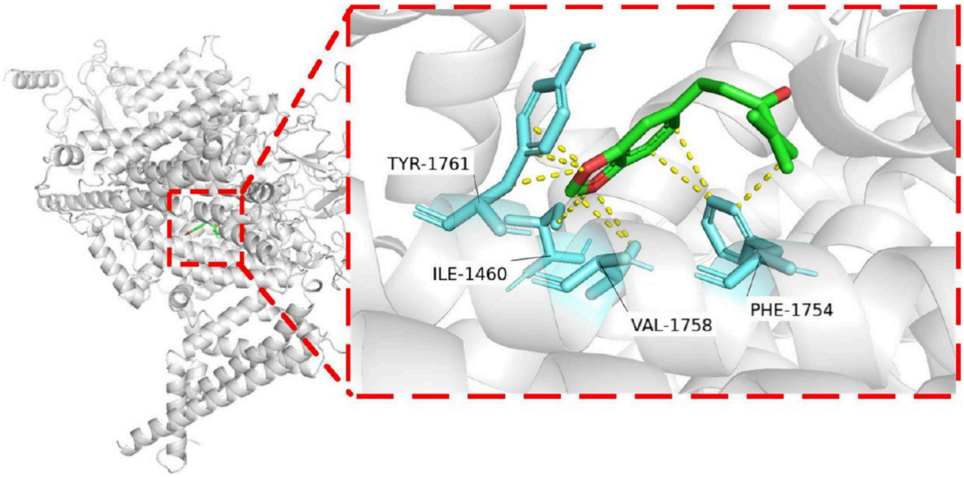

Figure 14. Visualization of molecular docking for DB09118 and Q9UQD0. Drug: DB09118, protein: Q9UQD0, affinity: −5.69 (kcal/mol).

5. CONCLUSION

In this manuscript, we proposed an end-to-end deep learning framework for DTI prediction based on multimodel representation learning and the gated multihead attention mechanism. DO-GMA provided a powerful and robust performance when predicting new DTIs. We foresee that DO-GMA can find potential targets for existing drugs and possible drugs targeting existing proteins and further facilitate drug repositioning and new drug discovery.

6. DATA AND SOFTWARE AVAILABILITY

All data are from publicly available resources. The BindingDB and BioSNAP data sets can be downloaded from https://github.com/peizhenbai/DrugBAN/tree/main/datasets. The C.elegans data sets can be downloaded from https://github.com/masashitsubaki/CPI_prediction/tree/master/dataset. The DrugBank data sets can be downloaded from https://github.com/zhaoqichang/GIFDTI/tree/main/data/DrugBank2021. The supplementary materials and source codes are available at https://github.com/plhhnu/DO-GMA.

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Lihong Peng: conceptualization, methodology, investigation, writing-original draft, writing-review and editing. Jiale Mao: methodology, investigation, software, formal analysis, data curation, visualization, writing-original draft, writing-review and editing. Guohua Huang: methodology, investigation, writing-original draft, writing-review and editing. Guosheng Han: supervision, funding acquisition. Xin Liu: methodology, formal analysis, writing-review and editing. Wen Liao: validation. Geng Tian: supervision, funding acquisition. Jialiang Yang: supervision, writing-review and editing, funding acquisition.

Funding

L.P. (Lihong Peng) was funded by the National Natural Science Foundation of China (Grant 61803151), Natural Science Foundation of Hunan Province (Grant 2023JJ50201), Scientific Research Project of Education Department of Hunan Province (Grant 23A0418), and Hunan Key Laboratory for Computation and Simulation in Science and Engineering, Xiangtan University. G.H. (Guohua Huang) was funded by the National Natural Science Foundation of China (Grant 62272310) and Scientific Research Foundation of Hunan Provincial Education Department (Grant 24A0701).

Notes

The authors declare no competing financial interest.

■ REFERENCES

(1) Wang, X.; Cheng, Y.; Yang, Y.; Yu, Y.; Li, F.; Peng, S. Multitask joint strategies of self-supervised representation learning on biomedical networks for drug discovery. *Nature Machine Intelligence* **2023**, *5*, 445−456.

(2) Zhou, Z.; Liao, Q.; Wei, J.; Zhuo, L.; Wu, X.; Fu, X.; Zou, Q. Revisiting drug−protein interaction prediction: a novel global−local perspective. *Bioinformatics* **2024**, *40*, btae271.

(3) Chen, X.; Huang, L. Computational model for drug research. *Briefings in Bioinformatics* **2024**, *25* (3), bbae158.

(4) Zhou, Y.; Zhang, Y.; Zhao, D.; Yu, X.; Shen, X.; Zhou, Y.; Wang, S.; Qiu, Y.; Chen, Y.; Zhu, F. TTD: Therapeutic Target Database

describing target druggability information. *Nucleic acids research* **2024**, *52* (D1), D1465−D1477.

(5) Yang, H.; Chen, Y.; Zuo, Y.; Deng, Z.; Pan, X.; Shen, H.-B.; Choi, K.-S.; Yu, D.-J. MINDG: A Drug-Target Interaction Prediction Method Based on an Integrated Learning Algorithm. *Bioinformatics* **2024**, *40* (4), btae147.

(6) Askr, H.; Elgeldawi, E.; Aboul Ella, H.; Elshaier, Y. A.; Gomaa, M. M.; Hassanien, A. E. Deep learning in drug discovery: an integrative review and future challenges. *Artificial Intelligence Review* **2023**, *56*, 5975−6037.

(7) Li, H.; Gao, Z.; Kang, L.; Zhang, H.; Yang, K.; Yu, K.; Luo, X.; Zhu, W.; Chen, K.; Shen, J.; Wang, X.; Jiang, H. TarFisDock: a web server for identifying drug targets with docking approach. *Nucleic acids research* **2006**, *34*, W219−W224.

(8) Junaidin, J.; Lestari, D.; Kurniawan, M. F.; Khairul Ikram, N. K. Ligand-based pharmacophore modeling, molecular docking, and molecular dynamic studies of HMG-CoA reductase inhibitors. *Informatics in Medicine Unlocked* **2022**, *32*, 101063.

(9) He, Q.; Liu, C.; Wang, X.; Rong, K.; Zhu, M.; Duan, L.; Zheng, P.; Mi, Y. Exploring the mechanism of curcumin in the treatment of colon cancer based on network pharmacology and molecular docking. *Frontiers in pharmacology* **2023**, *14*, 1102581.

(10) Babu, S.; Nagarajan, S. K.; Sathish, S.; Negi, V. S.; Sohn, H.; Madhavan, T. Identification of potent and selective JAK1 lead compounds through ligand-based drug design approaches. *Frontiers in Pharmacology* **2022**, *13*, 837369.

(11) Estrada, A. K.; Mendez-Alvarez, D.; Juarez-Saldivar, A.; Lara-Ramirez, E. E.; Martinez-Vazquez, A. V.; Villalobos-Rocha, J. C.; Palos, I.; Ortiz-Perez, E.; Rivera, G. Ligand-based and structure-based virtual screening of new sodium glucose cotransporter type 2 inhibitors. *Medicinal Chemistry* **2023**, *19*, 1049−1060.

(12) Hossain, M. A.; Sultana, S. In silico analysis of correlation between physicochemical properties of ligands and binding affinity to protein targets. *ChemRxiv* **2023**, 1.

(13) Shareef, U.; Altaf, A.; Zargaham, M. K.; Bhatti, R.; Ibrahim, A.; Zahid, M. A. Ligand Based Pharmacophore Modeling, Virtual Screening, Molecular Docking Molecular Dynamic simulation and In-silico ADMET Studies for the Discovery of Potential BACE-1 Inhibitors. *Research Square* **2023**, 1.

(14) Sajadi, S. Z.; Zare Chahooki, M. A.; Tavakol, M.; Gharaghani, S. Matrix factorization with denoising autoencoders for prediction of drug−target interactions. *Molecular Diversity* **2023**, *27*, 1333−1343.

(15) Peng, L.; Huang, L.; Su, Q.; Tian, G.; Chen, M.; Han, G. LDA-VGHB: identifying potential lncRNA−disease associations with singular value decomposition, variational graph auto-encoder and heterogeneous Newton boosting machine. *Briefings in Bioinformatics* **2023**, *25* (1), bbad466.

(16) Peng, L.; Xiong, W.; Han, C.; Li, Z.; Chen, X. CellDialog: A Computational Framework for Ligand-receptor-mediated Cell-cell Communication Analysis III. *IEEE Journal of Biomedical and Health Informatics* **2024**, *28* (1), 580.

(17) Peng, L.; Gao, P.; Xiong, W.; Li, Z.; Chen, X. Identifying potential ligand−receptor interactions based on gradient boosted neural network and interpretable boosting machine for intercellular communication analysis. *Computers in Biology and Medicine* **2024**, *171*, 108110.

(18) Qi, X.; Zhao, Y.; Qi, Z.; Hou, S.; Chen, J. Machine Learning Empowering Drug Discovery: Applications, Opportunities and Challenges. *Molecules* **2024**, *29*, 903.

(19) Tsubaki, M.; Tomii, K.; Sese, J. Compound-protein interaction prediction with end-to-end learning of neural networks for graphs and sequences. *Bioinformatics* **2019**, *35*, 309−318.

(20) Dehghan, A.; Abbasi, K.; Razzaghi, P.; Banadkuki, H.; Gharaghani, S. CCL-DTI: contributing the contrastive loss in drug−target interaction prediction. *BMC bioinformatics* **2024**, *25*, 48.

(21) Cai, D.; Lam, W. Graph transformer for graph-to-sequence learning. *Proceedings of the AAAI conference on artificial intelligence* **2020**, *34*, 7464−7471.

(22) Wang, T.; Wang, R.; Wei, L. AttenSyn: an attention-based deep graph neural network for anticancer synergistic drug combination prediction. *J. Chem. Inf. Model.* **2024**, *64*, 2854−2862.

(23) Meng, Y.; Wang, Y.; Xu, J.; Lu, C.; Tang, X.; Peng, T.; Zhang, B.; Tian, G.; Yang, J. Drug repositioning based on weighted local information augmented graph neural network. *Briefings in Bioinformatics* **2023**, *25* (1), bbad431.

(24) Peng, L.; Yang, C.; Yang, J.; Tu, Y.; Yu, Q.; Li, Z.; Chen, M.; Liang, W. Drug repositioning via Multi-view Representation Learning with Heterogeneous Graph Neural Network. *IEEE Journal of Biomedical and Health Informatics* **2024**, 1.

(25) Zhao, M.; Yuan, M.; Yang, Y.; Xu, S. X. CPGL: Prediction of compound-protein interaction by integrating graph attention network with long short-term memory neural network. *IEEE/ACM Transactions on Computational Biology and Bioinformatics* **2023**, *20*, 1935.

(26) Liu, S.; Wang, Y.; Deng, Y.; He, L.; Shao, B.; Yin, J.; Zheng, N.; Liu, T.-Y.; Wang, T. Improved drug−target interaction prediction with intermolecular graph transformer. *Briefings in Bioinformatics* **2022**, *23* (5), bbac162.

(27) Gao, M.; Zhang, D.; Chen, Y.; Zhang, Y.; Wang, Z.; Wang, X.; Li, S.; Guo, Y.; Webb, G. I.; Nguyen, A. T.N.; May, L.; Song, J. GraphormerDTI: a graph transformer-based approach for drug-target interaction prediction. *Computers in Biology and Medicine* **2024**, *173*, 108339.

(28) Shao, K.; Zhang, Z.; He, S.; Bo, X. DTIGCCN: Prediction of drug-target interactions based on GCN and CNN. In *IEEE 32nd International Conference on Tools with Artificial Intelligence (ICTAI)*; IEEE, 2020; pp 337−342.

(29) Liu, Z.; Chen, Q.; Lan, W.; Pan, H.; Hao, X.; Pan, S. GADTI: graph autoencoder approach for DTI prediction from heterogeneous network. *Frontiers in Genetics* **2021**, *12*, 650821.

(30) Zeng, P.; Zhang, B.; Liu, A.; Meng, Y.; Tang, X.; Yang, J.; Xu, J. Drug repositioning based on tripartite cross-network embedding and graph convolutional network. *Expert Systems with Applications* **2024**, *252*, 124152.

(31) Qi, H.; Yu, T.; Yu, W.; Liu, C. Drug−target affinity prediction with extended graph learning-convolutional networks. *BMC bioinformatics* **2024**, *25*, 75.

(32) Xie, Y.; Wang, X.; Wang, P.; Bi, X. A pseudo-label supervised graph fusion attention network for drug−target interaction prediction. *Expert Systems with Applications* **2025**, *259*, 125264.

(33) Zhao, T.; Hu, Y.; Valsdottir, L. R.; Zang, T.; Peng, J. Identifying drug−target interactions based on graph convolutional network and deep neural network. *Briefings in bioinformatics* **2021**, *22*, 2141−2150.

(34) Yu, D.; Liu, H.; Yao, S. Drug−target interaction prediction based on improved heterogeneous graph representation learning and feature projection classification. *Expert Systems with Applications* **2024**, *252*, 124289.

(35) Huang, K.; Xiao, C.; Glass, L. M.; Sun, J. MolTrans: molecular interaction transformer for drug−target interaction prediction. *Bioinformatics* **2021**, *37*, 830−836.

(36) Wang, G.; Zhang, X.; Pan, Z.; Rodríguez Patón, A.; Wang, S.; Song, T.; Gu, Y. Multi-transdti: transformer for drug−target interaction prediction based on simple universal dictionaries with multi-view strategy. *Biomolecules* **2022**, *12*, 644.

(37) Yin, Z.; Chen, Y.; Hao, Y.; Pandiyan, S.; Shao, J.; Wang, L. FOTF-CPI: A compound-protein interaction prediction transformer based on the fusion of optimal transport fragments. *iScience* **2024**, *27*, 108756.

(38) Monteiro, N. R.; Oliveira, J. L.; Arrais, J. P. DTITR: End-to-end drug−target binding affinity prediction with transformers. *Computers in Biology and Medicine* **2022**, *147*, 105772.

(39) Boezer, M.; Tavakol, M.; Sajadi, Z. FastDTI: drug-target interaction prediction using multimodality and transformers. In *Proceedings of the northern lights deep learning workshop 2023*; 2023.

(40) Liu, Y.; Xing, L.; Zhang, L.; Cai, H.; Guo, M. GEFormerDTA: drug target affinity prediction based on transformer graph for early fusion. *Sci. Rep.* **2024**, *14*, 7416.

(41) Monteiro, N. R.; Oliveira, J. L.; Arrais, J. P. TAG-DTA: Binding-region-guided strategy to predict drug-target affinity using transformers. *Expert Systems with Applications* **2024**, *238*, 122334.

(42) Zhang, L.; Wang, C.-C.; Chen, X. Predicting drug−target binding affinity through molecule representation block based on multi-head attention and skip connection. *Briefings in Bioinformatics* **2022**, *23*, bbac468.

(43) Li, M.; Lu, Z.; Wu, Y.; Li, Y. BACPI: a bi-directional attention neural network for compound−protein interaction and binding affinity prediction. *Bioinformatics* **2022**, *38*, 1995−2002.

(44) Hua, Y.; Song, X.; Feng, Z.; Wu, X. MFR-DTA: a multi-functional and robust model for predicting drug−target binding affinity and region. *Bioinformatics* **2023**, *39*, btad056.

(45) Nguyen, N.-Q.; Jang, G.; Kim, H.; Kang, J. Perceiver CPI: a nested cross-attention network for compound−protein interaction prediction. *Bioinformatics* **2023**, *39*, btac731.

(46) Bian, J.; Zhang, X.; Zhang, X.; Xu, D.; Wang, G. MCANet: shared-weight-based MultiheadCrossAttention network for drug−target interaction prediction. *Briefings in Bioinformatics* **2023**, *24*, bbad082.

(47) Tang, X.; Zhou, C.; Lu, C.; Meng, Y.; Xu, J.; Hu, X.; Tian, G.; Yang, J. Enhancing drug repositioning through local interactive learning with bilinear attention networks. *IEEE Journal of Biomedical and Health Informatics* **2024**, 1.

(48) Song, W.; Xu, L.; Han, C.; Tian, Z.; Zou, Q. Drug−target interaction predictions with multi-view similarity network fusion strategy and deep interactive attention mechanism. *Bioinformatics* **2024**, *40*, btae346.

(49) Peng, L.; Liu, X.; Yang, L.; Liu, L.; Bai, Z.; Chen, M.; Lu, X.; Nie, L. BINDTI: a bi-directional intention network for drug-target interaction identification based on attention mechanisms. *IEEE Journal of Biomedical and Health Informatics* **2024**, 1.

(50) Peng, L.; Bai, Z.; Liu, L.; Yang, L.; Liu, X.; Chen, M.; Chen, X. DTI-MvSCA: An anti-over-smoothing multi-view framework with negative sample selection for predicting drug-target interactions. *IEEE Journal of Biomedical and Health Informatics* **2025**, *29*, 711−723.

(51) Zhang, Y.; Liu, C.; Liu, M.; Liu, T.; Lin, H.; Huang, C.-B.; Ning, L. Attention is all you need: utilizing attention in AI-enabled drug discovery. *Briefings in bioinformatics* **2023**, *25*, bbad467.

(52) Bai, P.; Miljković, F.; John, B.; Lu, H. Interpretable bilinear attention network with domain adaptation improves drug−target prediction. *Nature Machine Intelligence* **2023**, *5*, 126−136.

(53) Wishart, D. S.; Knox, C.; Guo, A. C.; Shrivastava, S.; Hassanali, M.; Stothard, P.; Chang, Z.; Woolsey, J. DrugBank: a comprehensive resource for in silico drug discovery and exploration. *Nucleic Acids Res.* **2006**, *34*, D668−D672.

(54) Song, T.; Zhang, X.; Ding, M.; Rodriguez-Paton, A.; Wang, S.; Wang, G. DeepFusion: A deep learning based multi-scale feature fusion method for predicting drug-target interactions. *Methods* **2022**, *204*, 269−277.

(55) Liu, H.; Sun, J.; Guan, J.; Zheng, J.; Zhou, S. Improving compound−protein interaction prediction by building up highly credible negative samples. *Bioinformatics* **2015**, *31*, i221−i229.

(56) Gilson, M. K.; Liu, T.; Baitaluk, M.; Nicola, G.; Hwang, L.; Chong, J. BindingDB in 2015: A public database for medicinal chemistry, computational chemistry and systems pharmacology. *Nucleic Acids Res.* **2016**, *44*, D1045−D1053.

(57) Cao, J.; Li, Y.; Sun, M.; Chen, Y.; Lischinski, D.; Cohen-Or, D.; Chen, B.; Tu, C. Do-conv: Depthwise over-parameterized convolutional layer. *IEEE Transactions on Image Processing* **2022**, *31*, 3726−3736.

(58) Li, M.; Zhou, J.; Hu, J.; Fan, W.; Zhang, Y.; Gu, Y.; Karypis, G. Dgl-lifesci: An open-source toolkit for deep learning on graphs in life science. *ACS omega* **2021**, *6*, 27233−27238.

(59) Ilse, M.; Tomczak, J.; Welling, M. Attention-based deep multiple instance learning. In *Proceedings of the 35th International Conference on Machine Learning*; 2018; pp 2127−2136.

(60) Xiong, C.; Chen, H.; Sung, J. J.; King, I. Diagnose like a pathologist: Transformer-enabled hierarchical attention-guided multiple instance learning for whole slide image classification. *arXiv* **2023**, 1.

(61) Arar, M.; Shamir, A.; Bermano, A. H. Learned queries for efficient local attention. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE, 2022; pp 10841−10852.

(62) Han, D.; Ye, T.; Han, Y.; Xia, Z.; Song, S.; Huang, G. Agent attention: On the integration of softmax and linear attention. *arXiv* **2023**, 1.

(63) Tenenbaum, J. B.; Freeman, W. T. Separating style and content with bilinear models. *Neural computation* **2000**, *12*, 1247−1283.

(64) An, J.; Yeom, S.; Hong, T.; Jeong, K.; Lee, J.; Eardley, S.; Choi, J. Analysis of the impact of energy consumption data visualization using augmented reality on energy consumption and indoor environment quality. *Building and Environment* **2024**, *250*, 111177.

(65) Yan, L.; Wu, Q.; Li, X.; Xie, C.; Zhou, X.; Li, Y.; Shi, D.; Yu, L.; Zhang, T.; Tao, Y.; Yan, B.; Zhong, M.; Song, Z.; Yang, Y.; Huang, R. Neural Architecture Search with In-Memory Multiply−Accumulate and In-Memory Rank Based on Coating Layer Optimized C-Doped Ge2Sb2Te5 Phase Change Memory. *Adv. Funct. Mater.* **2024**, *34*, 2300458.

(66) Kirtane, M. V.; Bhandari, A.; Narang, P.; Santani, R. Cinnarizine: a contemporary review. *Indian Journal of Otolaryngology and Head & Neck Surgery* **2019**, *71*, 1060−1068.

(67) Russo, M.; De Rosa, M. A.; Calisi, D.; Consoli, S.; Evangelista, G.; Dono, F.; Santilli, M.; Granzotto, A.; Onofrj, M.; Sensi, S. L. Migraine pharmacological treatment and cognitive impairment: risks and benefits. *International Journal of Molecular Sciences* **2022**, *23*, 11418.

(68) Wu, Q.; Chen, C.; Liu, W.; Zhou, Y.; Weng, G.; Gu, Y. Network-based drug repurposing for potential stroke therapy. *Computational and Structural Biotechnology Journal* **2023**, *21*, 2809−2823.

(69) Arslan, Z.; Okuroğlu, E.; Şenol, H.; Türkmen, Z. 1-Benzhydryl-piperazine: Isolation, structure determination, and in silico studies for a novel potential narcotic agent detected in sports supplements. *Journal of Food Composition and Analysis* **2024**, *135*, 106682.

(70) Cerulli Irelli, E.; Cocchi, E.; Morano, A.; Gesche, J.; Caraballo, R. H.; Lattanzi, S.; Strigaro, G.; Rosati, E.; Catania, C.; Ferlazzo, E.; Casciato, S.; Di Gennaro, G.; Pizzanelli, C.; Giuliano, L.; Viola, V.; Mostacci, B.; Pignatta, P.; Fortunato, F.; Pulitano, P.; Panzini, C.; Gambardella, A.; Atalar, A. C.; Labate, A.; Operto, F. F.; Giallonardo, A. T.; Baykan, B. B.; Beier, C. P.; Di Bonaventura, C. Predictors of Seizure Recurrence in Women With Idiopathic Generalized Epilepsy Who Switch From Valproate to Another Medication. *Neurology* **2024**, *102*, No. e209222.

(71) Musto, E.; Liao, V. W.; Johannesen, K. M.; Fenger, C. D.; Lederer, D.; Kothur, K.; Fisk, K.; Bennetts, B.; Vrielynck, P.; Delaby, D.; et al. GABRA1-Related Disorders: From Genetic to Functional Pathways. *Ann. Neurol.* **2024**, *95*, 27−41.

(72) Alonso-Navarro, H.; García-Martín, E.; Agúndez, J. A.; Jiménez-Jiménez, F. J. Current and future neuropharmacological options for the treatment of essential tremor. *Current Neuropharmacology* **2020**, *18*, 518−537.

(73) Knox, C.; Wilson, M.; Klinger, C. M.; Franklin, M.; Oler, E.; Wilson, A.; Pon, A.; Cox, J.; Chin, N. E.; Strawbridge, S. A.; et al. DrugBank 6.0: the DrugBank knowledgebase for 2024. *Nucleic acids research* **2024**, *52*, D1265−D1275.

(74) Rondou, P.; Haegeman, G.; Van Craenenbroeck, K. The dopamine D4 receptor: biochemical and signalling properties. *Cellular and molecular life sciences* **2010**, *67*, 1971−1986.

(75) Morris, G. M.; Huey, R.; Lindstrom, W.; Sanner, M. F.; Belew, R. K.; Goodsell, D. S.; Olson, A. J. AutoDock4 and AutoDockTools4: Automated docking with selective receptor flexibility. *Journal of computational chemistry* **2009**, *30*, 2785−2791.

(76) DeLano, W. L. PyMOL: An open-source molecular graphics tool. *CCP4 Newsl. Protein Crystallogr.* **2002**, *40*, 82−92.

(77) Tu, Y.; Tan, L.; Tao, H.; Li, Y.; Liu, H. CETSA and thermal proteome profiling strategies for target identification and drug discovery of natural products. *Phytomedicine* **2023**, *116*, 154862.

(78) Luo, Y.; Zhao, X.; Zhou, J.; Yang, J.; Zhang, Y.; Kuang, W.; Peng, J.; Chen, L.; Zeng, J. A network integration approach for drug-target interaction prediction and computational drug repositioning from heterogeneous information. *Nat. Commun.* **2017**, *8*, 573.
(79) Öztürk, H.; Özgür, A.; Ozkirimli, E. DeepDTA: deep drug–target binding affinity prediction. *Bioinformatics* **2018**, *34*, i821–i829.
(80) Abramson, J.; Adler, J.; Dunger, J.; Evans, R.; Green, T.; Pritzel, A.; Ronneberger, O.; Willmore, L.; Ballard, A. J.; Bambrick, J.; et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **2024**, *630*, 493.

https://doi.org/10.1021/acs.jcim.4c02088
J. Chem. Inf. Model. 2025, 65, 1318–1337